S-TIM: A Spatio-Temporal Trust Inference Model for Malicious Node Detection in Low-Altitude UAV Networks

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Abstract: In a cooperative low-altitude UAV mission, nodes must choose which peers to keep, to send work to and to penalise, and a trust score matters only when it changes that choice. The choice can only be as good as the summary the trust aggregator is able to compute, so each node's behaviour, communication, energy, task-execution and reputation evidence has to be reduced to features that separate malicious from benign. Conventional trust aggregation reduces it to marginal window summaries, and across five rule-based comparators this leaves periodic on–off attackers at 0.0–38% detection. Two blind spots remain, and both belong to the representation rather than to the rules that consume it: marginal window summaries cannot express the intermittency of a scheduled attacker, and a method that scores one node at a time is blind to an attacker whose telemetry is indistinguishable from the benign population, at 2.1% detection. We instantiate the two missing capabilities in S-TIM, a supervised encoder over per-node marginal, temporal-shape and neighbourhood-relative features. Five scale-free shape statistics—autocorrelation excess, sign-change rate, quantile spread and sign-change-interval variation—separate a scheduled attacker from a noisy one without assuming an attack period, lifting the same models to at least 88%; five neighbourhood-relative statistics compare a node against the link conditions its neighbours actually experience, raising detection of the behaviourally-clean attacker above 83%. S-TIM exceeds the strongest rule-based comparator by +9.0 pp accuracy on paired seeds and leads every aggregate metric by +49.8 pp accuracy over the strongest rule adapter in the scenario that contains that attacker ($p < 0.0001$). The gain belongs to the representation: with the encoder, the input width and the early-stopping rule held fixed, no arrangement of the auxiliary heads changes the detection rates, and the same holds for alternative training objectives. The blocks are not free, and their costs are reported alongside the gains. The executable implementation is a supervised trust-stream pipeline, and the limits of what the evaluation supports are stated rather than left implicit.

Keywords: UAV networks; trust management; malicious node detection; task allocation; reproducible simulation

1 Introduction

UAV swarms used for inspection, sensing, and emergency response rely on nodes that remain inside the mission after admission. A valid identity is therefore not sufficient evidence of trustworthy behaviour. A compromised node can selectively forward traffic, falsify a task result, or alternate between benign and harmful behaviour while retaining the privileges granted by its earlier history. Dynamic topology and intermittent links make those behaviours difficult to distinguish from ordinary failures [1, 2].

Trust management addresses this problem by updating a node score from recent observations. Existing schemes use fuzzy fusion, anomaly profiles, provenance, or group decisions [7, 8, 10, 22]. Their scores, however, are often evaluated only as detector outputs. In a mission, the practical question is what changes after a node becomes risky: whether it may receive a sensitive task, how much traffic it may relay, and when it should be isolated. This study examines that narrower trust-to-allocation question.

We present S-TIM (Spatio-Temporal Trust Inference Model) as a spatio-temporal trust evaluation method and as a reproducible proof of concept. The implementation generates five trust-factor streams, fits a supervised encoder, and uses its risk output in a model-based task-allocation simulation. The released training procedure is a supervised one: the risk, action and value targets all come from the simulator's labels, and the encoder is fitted on held-out training seeds with a seed-level validation split.

The blindness described above is specific to attacks that live in a node's own behaviour, and it is

therefore silent about the attacker that matters most when the adversary knows the aggregator scores single nodes over time. Such a node emits well-formed telemetry, accumulates trust at the rate a benign node does, and can hold it indefinitely; every statistic a per-node aggregator computes is by construction the statistic a benign node in the same radio cell would produce. No amount of temporal detail can separate the two, because the information that separates them is not in the node. This second blind spot is spatial rather than temporal, and closing it requires a different kind of statistic: one that compares a node against the link conditions its neighbours actually experience.

The study asks three bounded questions. First, how should generated trust streams be summarised so that constant, dormant, and on–off attackers are all separable on held-out random seeds? Second, on the same evidence, what does an aggregator that scores one node at a time fail to see, and what does a neighbourhood-relative statistic add to it? Third, under the allocation rules encoded in the same simulator, does trust-guided selection improve a task-success proxy? The answer to the first question is that the summarisation step, not the aggregation rules built on it, is the binding constraint: the marginal window aggregates that conventional trust aggregation uses leave periodic on–off attackers at 0.0–38% detection across five rule-based comparators, and five scale-free shape statistics lift the proposed model to 88.0%, a +50.0 to +88.0 pp gap that is concentrated entirely on the one attack class the aggregators are structurally unable to express. We instantiate the finding in S-TIM, a supervised encoder over the marginal block plus the shape block, and show that the representation rather than the encoder is what closes the gap: switching the auxiliary supervision on and off moves nothing, because a probe shows the auxiliary target is already 96.7% recoverable from the marginal features, and so does replacing the training objective with a non-saturating one. A paired duty-cycle sweep then shows how far this goes—below a 20% active fraction the rule-based aggregator is functionally blind, identifying none of 150 pooled on–off nodes at a 5% duty, while the shape-aware method still recovers a seventh of them. The answer to the second question is that the per-node view is structurally blind to an attacker whose behaviour is clean, and that the missing information is recoverable: with the marginal and shape blocks alone such a node is detected at 2.1%, while five neighbourhood-relative statistics raise the same model to 83.6% on the same seeds, a +81.5 pp paired gain ($t = 63.7$, $p < 0.0001$). The reason is structural rather than architectural: the information that separates such a node from a benign one is not in the node at all, so it has to come from the link conditions the neighbours actually experience. The answer to the third is suggestive rather than causal because the task simulator contains mechanism-specific rules. The limits of what the evaluation supports are stated rather than demonstrated by a second scenario.

Problem statement. This paper addresses a trust-layer decision problem for fixed identities: how should a risk estimate derived from noisy multi-factor evidence affect a subsequent task-allocation decision? The empirical scope is a generated trust-stream benchmark. It does not cover admission control, Sybil resistance, real-time distributed execution, flight deployment, or the validation of absolute position integrity; the spatial result below is a statement about what a trust aggregator can and cannot resolve from the trust stream itself, not a position-verification scheme.

Figure 1 gives the executed pipeline, and Figure 2 gives the system design the method is written for, marking the parts the implementation leaves open. The contributions are consequently limited to the following:

- a documented generator for behavioural, communication, energy, task, and historical trust streams under three temporal attack patterns and one spatially-inconsistent pattern, in which a subset of the malicious population reports telemetry drawn from the benign distribution while standing in an airspace cell whose radio field differs from the one it reports;
- a characterisation of a second, independent blind spot, which no amount of temporal detail can close: an attacker whose behaviour is clean is invisible to every method that scores one node at a time. With the marginal and shape blocks alone it is detected at 2.1%. A five-dimensional neighbourhood-relative block raises the same model to 83.6% on the same seeds, a +81.5 pp paired gain ($t = 63.7$, $p < 0.0001$), while the behaviour channels of that attacker remain indistinguishable from benign ones, with effect sizes below 0.04 against the benign population against 1.99 for the neighbourhood-relative statistics;
- the headline detector results, measured on paired evaluation seeds: S-TIM exceeds the strongest rule-based comparator by +9.0 pp accuracy and +15.0 pp F-score ($p < 10^{-6}$), and the method-group analysis shows why, with on–off detection rising from 0.0–38% across five rule-based aggregators to 88.0% for the proposed model. In the spatially inconsistent scenario, in which the comparison is not compressed by a perfectly separable training split, S-TIM leads every aggregate metric with 93.8% accuracy, 89.4% F-score and 88.9% task-success rate (TSR), against 39.7–44.0%, 44.9–45.5% and

74.5–75.9% for the rule adapters, a margin over the strongest of them on each metric of +49.8 pp accuracy, +43.9 pp F-score and +13.0 pp TSR ($p < 0.0001$); averaged equally over the four attack classes rather than over the 800 nodes, of which 560 are benign, it reaches 89.0% against 85.1% for the strongest rule adapter, so it leads all five baselines on that summary as well;

- a controlled identification of the mechanism, with a reproducible supervised encoder on disjoint training and evaluation seeds: with the encoder, the input width and the early-stopping rule held fixed, switching every auxiliary loss off, supervising only the attack-modality head, or supervising all four changes no detection metric, and an information probe shows why, since the auxiliary target is already 96.7% recoverable from the marginal features alone. The block ablations of the same model locate the gain in the representation: removing the shape block lowers on–off detection from 88.0% to 72.0%, and removing the neighbourhood block lowers the behaviourally-clean attacker from 83.6% to 2.1%, so the gain is a property of the evidence the model is given rather than of the aggregation rules the evidence is meant to feed;

The remainder of this paper reviews related work, states the system design and the part of it that is implemented and evaluated, reports the trust-stream experiments, and closes with limitations.

2 Related Work

Trust evaluation for UAV networks now includes reputation models, evidence fusion, blockchain support, and learning-based designs. Yet low-altitude UAV studies rarely integrate zero-trust verification, distributed trust learning, uncertainty aggregation, and mission incentives. We review three related areas: UAV trust management, position-aware trust and integrity monitoring, and zero-trust assessment.

2.1 Trust Management in UAV and FANET Systems

Recent surveys show that trust management has become a core security primitive in FANETs, where dynamic topology, intermittent links, and resource limits make purely identity-based protection insufficient [1–5]. Existing schemes fall into three categories: behaviour-centric trust analytics, distributed or blockchain-supported trust, and learning-enhanced trust evaluation.

Behaviour-centric trust evaluation widely uses fuzzy and evidence-fusion methods to combine heterogeneous observations. Benfriha et al. proposed FUBA, which uses fuzzy logic to integrate indicators such as RSSI, packet delivery ratio, delay, battery level, and humidity to identify insider threats in FANETs [8]. Li et al. developed an adaptive multi-granularity trust management scheme for UAV visual sensor security under adversarial attacks, where coarse- and fine-grained evidence is fused to improve robustness against known and unknown threats [9]. Akram et al. introduced GALTrust, which combines generative adversarial learning with type-2 fuzzy logic to improve adaptability to emerging malicious behaviours in spatial crowdsourcing drone services [10]. More recent federated-security studies also move trust assessment towards collaborative anomaly detection, including FL-based intrusion detection for FANETs, multimodal swarm anomaly detection, and communication-efficient FL for resource-constrained UAV-IoT systems [11–13]. Moving to task offloading, Kumar et al. evaluate trust for UAV-as-a-Service by combining residual energy, achievable data rate, and a reputation score that is transferred between a task and a service UAV through a Rubinstein bargaining game [23]. Their trust value is an energy gate multiplied by a weighted reputation and rate term, so it expresses resource adequacy and prior success rather than adversarial intent, and the paper reports a small improvement in delay, energy, and failure rate with no adversary in the loop. These studies significantly enrich trust evidence modelling. Their remaining limitation for the problem considered here is not that they cannot detect insider behaviour at all, but that fuzzy, profile-based, or task-specific fusion can become sensitive to distribution shifts when the malicious population is dense and attack timing is deliberately evasive.

A second line of work improves trust persistence and auditability through distributed storage or hierarchical trust exchange. Ge et al. developed a provenance-aware distributed trust model for resilient UAV networks, highlighting the value of peer-to-peer evidence tracing in hostile environments [7]. Zuo et al. proposed a hierarchical blockchain-based trust measurement method for drone clusters to address trust inheritance, cross-cluster re-evaluation, and historical trust verification under dynamic node entry and exit [6]. Akram et al. proposed TMIoDT, which integrates digital twins, blockchain, and deep-learning-based intrusion detection to secure Open RAN-based Internet of Drone Things applications [16]. Closely related

secure-FL frameworks have further explored malicious-node detection and trusted collaboration through blockchain-supported service provisioning, privacy-preserving IoDT security, and hierarchical federated aggregation [17–20]. Xu et al. pair a Bayesian trust assessment for quantum federated learning with Stackelberg incentives for the same UAV-assisted setting [21]. The two parts are separable: the incentive side is a full game-theoretic construction, whereas the trust degree itself is the posterior mean of a Beta–Binomial model, that is, a smoothed cumulative proportion of high-quality model updates. Their reported neutral device, which alternates between good and bad updates, therefore settles at an intermediate trust value rather than converging to a malicious verdict—an outcome the aggregator implies by construction rather than by choice. More recently, Hu et al. proposed GDM-DTM for malicious-node detection in low-altitude UAV networks by combining group decision making with dynamic trust management [22]. Its attacker model is a single message-tampering and false-information scheme, in which malicious nodes alter the attributes they forward and broadcast fabricated messages, and it recognises no periodic behaviour, so a window summary remains adequate against it and the question this manuscript asks is not posed. Its consistency evaluation also aggregates the neighbours' trust reports about a node, and therefore depends on the willingness of those neighbours to report, whereas the comparison used here is between a node's own channels and the radio conditions its neighbours actually experience, which requires no report. These schemes improve distributed credibility, trust persistence, and malicious-node detection. For our setting, the central gap is that the trust state is usually treated as an assessment output rather than as the basis for learned reward, penalty, and task-allocation decisions.

Three problems remain open in that literature for low-altitude UAVs, and they are the ones this paper addresses.

The first is accurate trust assessment under high malicious-node ratios: fuzzy, profile-based, and group-decision mechanisms can lose discrimination when the malicious population substantially shifts local trust distributions, leading to poor mission continuity.

The second is temporal and multi-factor evasion: dormant and on–off attackers can exploit historical trust accumulation, while multi-factor manipulation requires deciding which behavioural, communication, energy, task, and history evidence remains reliable at each update. This problem is not open. On-off-aware trust management is established, from the model that defends on–off attacks directly [25] through its distributed variant [26] and the smart-middleware detector [27], to the demonstration that cumulative and exponentially weighted moving averages do not work well under on–off attacks and the asymmetric weighting that repairs them [28], and to the recent model that criticises sliding-window aggregation for being unable to select an appropriate window length [29]. What these mechanisms share is that they intervene on the aggregated trust value or on the aggregation that produces it—they re-weight it, re-window it, mitigate the damage it does, or index its fragmentation [24]—and an index applied afterwards inherits whatever the underlying aggregator has already smoothed away. The question asked here is upstream of all of them, and it is the one they leave open: whether the information an intermittent attacker destroys can be recovered at all from the summaries a fixed-window aggregator is defined to produce. Our answer is that it cannot be recovered from those summaries, and that the recovery requires different statistics over each channel rather than a different weighting, a different window, or an index applied afterwards.

The third is spatially inconsistent attackers: an attacker whose telemetry is drawn from the benign population is invisible to every aggregator that scores one node at a time, because the statistics such an aggregator computes are by construction the statistics a benign node in the same radio cell produces. The missing information is not in the node, so no amount of temporal detail recovers it, and the position-integrity literature that does address this class of attacker relies on an absolute reference that a swarm in infrastructure-scarce conditions does not have. What is needed is a statistic that compares a node against the conditions its neighbours actually experience, which is a question about what a trust aggregator must represent rather than about which detection rule to apply.

2.2 Position-Aware Trust and Integrity Monitoring in UAV Swarms

A separate line of work attacks the problem from the position side. A UAV that forges its reported location can steer a swarm, corrupt cooperative localization, and carry a payload to a forbidden region while every behavioural statistic it reports remains plausible. This is an active literature. Zhong et al. enhance the resilience of UAV networks against GPS spoofing through a Byzantine distributed detection algorithm [34], and Tariq and Ahamed Ahanger address BeiDou spoofing in decentralized swarms with

multi-model fusion [38]; Dang et al. detect spoofing on cellular-connected UAVs with a deep ensemble [35], and Zhao et al. survey the multi-source-fusion detectors that the field has accumulated [37]. On the measurement side, Fang, Han, and Schotten give an information-theoretic account of the performance and security trade-off in trustworthy UAV cooperative localization [36], and Keshavarz et al. build a real-time trust-monitoring framework for a UAV network [39].

Two features of that literature matter for the present question. First, most of these mechanisms need a reference: an absolute position estimate, a trusted reference station, or a second sensor whose own integrity is assumed. The anchor-free alternative is comparatively young. Vijay et al. propose range-based multi-robot integrity monitoring that is explicitly anchor-free, detecting compromised or faulty robots from the consistency of inter-robot ranges rather than from an absolute reference [33]. Second, these mechanisms operate on the positioning stack, whereas trust management operates on an evidence stream that the positioning stack never sees. The question they leave open is what a trust aggregator can resolve when the only evidence available is the per-node trust stream and no trusted anchor exists at all. Cluster-level trust management in drone swarms [6] inherits the same limitation: it exchanges and hierarchically aggregates trust, but every node it evaluates is judged against its own history and its own report.

The attacker considered here lies precisely in that gap. It does not forge a position and does not need to: it reports well-formed telemetry consistent with the benign population and stands in an airspace cell whose radio conditions differ from the ones it reports, so that only its neighbours disagree with it. No absolute reference is required to detect it, and no absolute reference is available. The question asked here is therefore upstream of the integrity-monitoring literature in the same sense that it is upstream of the on–off-aware trust literature: not which detection rule is better once the right evidence has been identified, but what evidence has to exist in the first place for any rule to work.

2.3 Zero-Trust and Continuous Trust Assessment

Zero trust shifts security from perimeter-based assumptions to continuous, resource-centric verification. The NIST zero-trust architecture defines this paradigm through the principle of "never trust, always verify" and replaces implicit perimeter-based trust with continuous, context-aware verification [15]. Building on this idea, Wang et al. proposed a zero-trust-driven collaborative intrusion detection framework for IoT that uses continuous trust assessment to support distributed security decisions [30]. In the UAV and 6G context, Al Ridhawi and Aloqaily integrated zero trust, digital twins, federated learning, and blockchain into a UAV-enabled 6G framework, highlighting the broader relevance of decentralized trust enforcement in highly dynamic aerial networks [31]. In the UAV authentication domain, Wang et al. introduced a context-aware trust evaluation model for malicious UAV detection during authentication, showing that trust can also be injected into the authentication stage rather than being treated purely as a post hoc reputation signal [14].

Zero trust is particularly relevant to low-altitude UAV swarms: node roles may change during flight, links may appear and disappear rapidly, and local observations may be incomplete or noisy. Trust decisions cannot be based solely on prior membership or static credentials, but must be continuously updated from current observations, device status, communication behaviour, and mission context. Therefore, trust decisions should be dynamic, uncertainty-aware, and resource-conscious rather than static or binary.

These studies move trust evaluation closer to continuous verification. Nevertheless, the current zero-trust literature still focuses mainly on authentication, access control, or intrusion detection. It does not fully address how trust should evolve online in low-altitude UAV swarms when heterogeneous evidence from behaviour, communication quality, task execution, energy state, and historical interactions must be fused under strict resource constraints. In particular, uncertainty-aware aggregation and decentralized incentive adaptation remain underexplored in zero-trust UAV settings.

3 System Model

We consider a low-altitude UAV swarm executing a cooperative mission in an infrastructure-scarce environment. This section states the system the method is designed for, and marks which parts of it the implementation and the evaluation cover.

3.1 Problem Definition

At each update interval, the system receives partial and noisy observations of each UAV's behaviour, communication quality, energy state, task execution, and historical reputation. The objective is to infer a trust state that reflects the UAV's current behaviour and then use that state to regulate future participation through rewards, penalties, and task-allocation preferences. Formally, for each UAV U_k, S-TIM must output a global trust score $\mathcal{T}_k(t)$, a trust class $C_k(t)$, and an incentive action $\pi_{jk}(t)$ for the local observer j that supervises U_k:

$$(\mathcal{T}_k(t), C_k(t), \pi_{jk}(t)) = \Pi_\theta(o_j(t), \mathcal{H}_k(t)) \,, \tag{1}$$

where $o_j(t)$ is the partial observation available to observer j, $\mathcal{H}_k(t)$ is the recent trust-evidence history of node k, and Π_θ is the learned trust–incentive policy. The conceptual objective is to improve both malicious-node detection and mission continuity while respecting update-cycle resource budgets:

$$\max_{\Pi_\theta} \; \mathbb{E}[\lambda_1 F_1 + \lambda_2 \mathrm{TSR} - \lambda_3 C_{\mathrm{comm}} - \lambda_4 T_{\mathrm{eval}}] \,, \tag{2}$$

subject to bounded trust scores and an incentive budget. This is a conceptual objective only. The empirical model minimizes the supervised loss in Section 3.5.3; it does not optimise F_1, TSR, communication cost, or latency jointly.

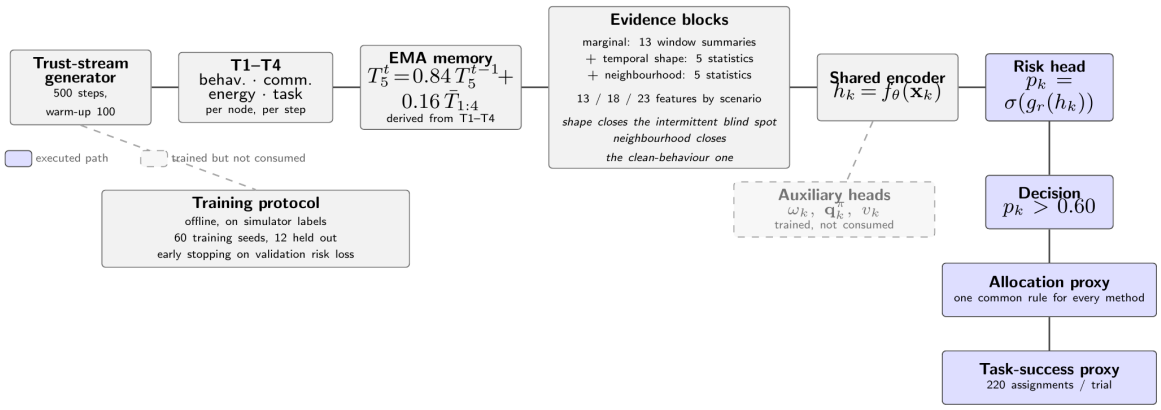


Figure 1: The executed pipeline. Every box on the solid path is run by the released implementation: the trust-stream generator, the three evidence blocks, the shared encoder, the risk head, the fixed decision threshold, the allocation proxy and the task-success proxy. The auxiliary heads are trained but not consumed by inference. The implementation runs one supervised network over per-node features; the pipeline is a feed-forward pass from the generated streams to the task-success proxy, with no closed state-feedback loop.

3.2 UAV Network Model

We model the low-altitude UAV network as a dynamic ad hoc network of N heterogeneous UAV nodes, $\mathcal{U} = \{U_1, U_2, \ldots, U_N\}$. Each node U_i is characterised by a tuple $\langle \mathbf{p}_i(t), v_i(t), E_i(t), C_i \rangle$, where $\mathbf{p}_i(t) \in \mathbb{R}^2$ is the 2D position at time t, $v_i(t) \in [v_{\min}, v_{\max}]$ is the velocity, $E_i(t) \in [0, E_i^{\max}]$ is the residual energy, and C_i denotes the computational capacity (CPU cycles/s). Nodes may enter or leave the swarm during the mission according to operational requirements.

The communication topology is a time-varying undirected graph $G(t) = (\mathcal{U}, \mathcal{E}(t))$, where an edge $(i, j) \in \mathcal{E}(t)$ exists iff the Euclidean distance satisfies $d_{ij}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2 \le R_{\mathrm{comm}}$, with $R_{\mathrm{comm}} =$ 300 m (Table 2). Each link (i, j) has a time-varying quality characterised by packet delivery ratio $\mathrm{PDR}_{ij}(t)$, received signal strength $\mathrm{RSSI}_{ij}(t)$, and one-way latency $L_{ij}(t)$. The network is *infrastructure-scarce*: no fixed base stations or ground controllers are assumed to be continuously reachable; all trust computation and decision-making must execute on-board or via peer-to-peer exchange.

3.3 Threat Model

We use a fixed-identity synthetic adversary model. It is not adaptive to the learned parameters.

Adversary capabilities. A subset of nodes $\mathcal{M} \subset \mathcal{U}$ ($|\mathcal{M}|/N \in [0.05, 0.50]$ in our experiments) is controlled by an adversary. Each malicious node $U_m \in \mathcal{M}$ can independently manipulate any subset of the five trust dimensions by:

- **T1 (Behavioral)**: deviating from planned flight paths, skipping scheduled heartbeats, or responding inconsistently to coordination messages;
- **T2 (Communication)**: performing selective forwarding (dropping a fraction $p_{\text{drop}} \in [0.3, 1.0]$ of packets), injecting spurious traffic, or manipulating transmission power to create asymmetric links;
- **T3 (Energy)**: exhibiting anomalous consumption patterns (e.g., sudden depletion through jamming, or unusually low consumption by remaining idle during active mission phases);
- **T4 (Task)**: returning fabricated or degraded task results (e.g., falsified sensor readings, partial payload delivery);
- **T5 (Historical)**: this dimension is not directly manipulable by the adversary—it is an accumulated reputation signal maintained by S-TIM—but its update is indirectly affected when the adversary manipulates T1–T4.

Attack strategies. We consider four canonical attack patterns, three temporal and one spatial, that cover the spectrum from naive to evasive adversaries:

(i) **Constant attack**: the malicious node exhibits adversarial behaviour from $t = 0$ and maintains it throughout the mission. This models unsophisticated attackers and serves as a lower bound on detection difficulty.

(ii) **Dormant attack**: the malicious node behaves honestly during an initial warm-up period ($t \leq T_{\text{warm}}$), accumulating trust, then switches to adversarial behaviour. This models adversaries that attempt to evade initial screening.

(iii) **On-off attack**: the malicious node toggles between honest and adversarial behaviour with period T_{toggle}, attempting to keep its trust score near the classification boundary where decisions are most uncertain.

(iv) **Spatially inconsistent attack**: the malicious node reports channels drawn from the benign population, so that every behavioural statistic computed on the node is statistically indistinguishable from a benign node's, and it is placed in an airspace cell whose radio field differs from the one it reports. Its neighbours carry the local field offset that it does not. The attack is invisible to any aggregator that scores one node, and visible only to a statistic that compares a node against the conditions its neighbourhood actually experiences.

In mixed-attack experiments, the temporal patterns are assigned cyclically across malicious nodes, and the spatially inconsistent pattern is applied to a configurable fraction of them. The implementation does not implement colluding peer reports, Sybil identities, an adaptive attacker, or training-data poisoning.

Adversary knowledge. The generated attacker knows neither the fitted parameters nor the decision threshold. Because it does not respond to model outputs, the experiments measure generalisation across random seeds from one generator, not robustness to a strategic white-box adversary.

Spatial field. The spatial scenario superimposes a smooth channel-quality field over the airspace, interpolated from a coarse random grid at the node positions. Two nodes within communication range therefore observe correlated conditions, which is what gives a neighbourhood any information to offer: the absolute level of a channel now says something about where a node is as well as about what it is doing. The field is applied as a common mode over all four evidence channels rather than to a subset of them, so that the cross-channel gap features remain field-invariant and continue to measure what they are meant to measure. The benign reference is shifted downward by a configurable amount so that the field's excursions clip symmetrically against the lower and upper bounds; without that shift the clipped benign mean drifts away from the reference, and a node that carries no field offset would become visible to a purely behavioural method for a reason that has nothing to do with its behaviour. A fraction of the malicious population is then placed in a cell whose field value differs from the one it reports, and the field offset that node would carry is removed from its own channels while its neighbours keep theirs.

Both corrections are diagnostics rather than tuning: each was found by a script in the released analysis directory, and each was verified to leave the published configuration untouched.

Design implications. The generator mainly tests whether temporal summaries of T1 and T4 separate the three temporal attack profiles from benign nodes, and whether a neighbourhood-relative statistic separates the fourth. It does not test majority collusion or Byzantine aggregation, and the spatially inconsistent attacker is generated by the same machinery rather than by an adversary that adapts to the model. Any deployment would still require cryptographic identity binding and an independently validated observation pipeline.

3.4 Multi-factor Trust Evaluation Mechanism

S-TIM represents credibility with five trust dimensions. They are not statistically independent, and the present study does not support an "exponentially harder" evasion claim. T5 is explicitly computed from the other four dimensions and is therefore a memory term, not a separate sensor.

3.4.1 Trust Factor Decomposition

T1 denotes behavioural consistency, T2 communication reliability, T3 energy behaviour, and T4 task quality. T5 is an exponentially smoothed mean of T1–T4. Equations 3–6 describe how these quantities could be obtained from telemetry. In the main experiment, however, the implementation samples normalised T1–T4 values directly from attack-dependent distributions and computes T5 from those samples; it does not simulate the raw packet, position, energy, or task variables in these equations.

The trust evaluation is decomposed into multiple trust branches, each corresponding to a distinct trust factor. Let the overall trust score of UAV node i at time t be denoted as $T_i(t) \in [0, 10]$, which is a function of five fundamental trust dimensions (the formal aggregation mechanism is presented in Section 3.4.2), where each trust branch $T_i^{(m)}(t) \in [0, 1]$ is defined as follows:

- **T1: Behavioral Trust** $T_i^{(1)}(t)$. This factor evaluates operational consistency, including adherence to flight paths, communication regularity, and response reliability:

$$T_i^{(1)}(t) = \frac{1}{N_b} \sum_{j=1}^{N_b} \mathbb{I}\Big(a_{i,j}(t) = a_{i,j}^{expected}(t)\Big) \tag{3}$$

  where $\mathbb{I}(\cdot)$ is the indicator function, $a_{i,j}(t)$ represents the actual action in behaviour category j, and $a_{i,j}^{expected}(t)$ is the expected behaviour. This yields $T_i^{(1)}(t) \in [0, 1]$.

- **T2: Communication Trust** $T_i^{(2)}(t)$. This factor assesses link quality and stability through packet delivery ratio, signal strength, and latency:

$$T_i^{(2)}(t) = w_p \cdot PDR_i(t) + w_s \cdot \frac{RSSI_i(t)}{RSSI_{max}} + w_l \cdot \left(1 - \frac{L_i(t)}{L_{max}}\right) \tag{4}$$

  where $PDR_i(t)$, $RSSI_i(t)/RSSI_{max}$, and $1 - L_i(t)/L_{max}$ are each normalised to $[0, 1]$, and $w_p = 0.4, w_s = 0.3, w_l = 0.3$ are weighting coefficients with $w_p + w_s + w_l = 1$. Thus $T_i^{(2)}(t) \in [0, 1]$.

- **T3: Energy Trust** $T_i^{(3)}(t)$. This factor monitors energy-consumption patterns and remaining battery life to detect anomalies indicative of malfunction or compromise:

$$T_i^{(3)}(t) = \frac{E_{remaining}(t)}{E_{initial}} \cdot \exp\left(-\alpha \cdot \left|C_i(t) - \bar{C}(t)\right|\right) \tag{5}$$

  where $\frac{E_{remaining}(t)}{E_{initial}} \in [0, 1]$ and the exponential term with $\alpha = 0.5$ is also bounded in $[0, 1]$, yielding $T_i^{(3)}(t) \in [0, 1]$.

- **T4: Task Execution Trust** $T_i^{(4)}(t)$. This factor measures the success rate and quality of task completion, such as sensing, data relaying, or payload delivery:

$$T_i^{(4)}(t) = \frac{1}{N_t} \sum_{k=1}^{N_t} \left(\frac{Q_{i,k}(t)}{Q_k^{max}} \cdot \mathbb{I}(\text{task_success}) \right) \tag{6}$$

  where each term is normalised to $[0, 1]$, so $T_i^{(4)}(t) \in [0, 1]$.

- **T5: Historical Trust** $T_i^{(5)}(t)$. This factor incorporates long-term reputation scores and past behaviour records to preserve temporal continuity in trust evaluation:

$$T_i^{(5)}(t) = \beta \cdot T_i^{(5)}(t-1) + (1-\beta) \cdot T_i^{inst}(t) \tag{7}$$

  where both the historical and instantaneous components are in $[0, 1]$, and the smoothing factor $\beta = 0.7$. Thus $T_i^{(5)}(t) \in [0, 1]$.

3.4.2 Multi-factor Trust Aggregation

Each trust branch independently computes a sub-score $T_i^{(m)}(t) \in [0, 1]$ based on real-time data streams and historical logs. These sub-scores are then aggregated into a global trust score $T_i(t) \in [0, 10]$ through a weighted fusion mechanism:

$$T_i(t) = 10 \cdot \sum_{m=1}^{5} \omega_m(t) \cdot T_i^{(m)}(t) \tag{8}$$

where the weights sum to one. The network learns an auxiliary weight head, but the released classifier computes risk directly from 13 summary features and does not use Eq. 8 to obtain its prediction.

3.4.3 Cross-Validation Mechanism

The multi-factor design ensures that no single point of failure or manipulation can compromise the overall trust decision. To enhance robustness, a **cross-validation mechanism** is implemented between trust branches. For any pair of trust factors m and n, the consistency metric is defined as:

$$\Gamma_{m,n}(t) = \left| T_i^{(m)}(t) - T_i^{(n)}(t) \right| \tag{9}$$

If $\Gamma_{m,n}(t) > \gamma_{\text{th}}$, where the cross-validation threshold $\gamma_{\text{th}} = 0.3$ (determined via sensitivity analysis over $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ with 0.3 yielding the best F-score on a validation subset), a potential inconsistency is flagged, triggering deeper inspection. For instance, even if a UAV exhibits normal communication behaviour ($T_i^{(2)}(t)$ high), but shows abnormal energy depletion ($T_i^{(3)}(t)$ low), the system can still flag it as potentially compromised. This layered verification keeps trust continuously evaluated rather than assumed from any single evidence channel.

Furthermore, the framework supports **anomaly detection** through statistical analysis of trust factor deviations:

$$\mathcal{A}_i(t) = \mathbb{I}\left(\sum_{m=1}^{5} \left(\frac{T_i^{(m)}(t) - \mu_m(t)}{\sigma_m(t)} \right)^2 > \chi^2_{\alpha,5} \right) \tag{10}$$

where $\mu_m(t)$ and $\sigma_m(t)$ are the mean and standard deviation of trust factor m across the swarm, $\chi^2_{\alpha,5}$ is the chi-squared threshold with 5 degrees of freedom at significance level $\alpha = 0.05$. This redundancy and diversity in trust evidence form the foundation for robust and adaptive trust management in UAV swarms, enabling the detection of sophisticated attacks that may manipulate multiple dimensions simultaneously.

3.5 Spatio-Temporal Risk Model and Allocation Proxy

The intended architecture divides observation across local observers and forwards their estimates to a decision node. The executable artifact is simpler. It applies one shared supervised network independently to each node's diagnostic feature vector. Observer assignments do not enter the feature extractor, and neither confidence-weighted fusion nor budget projection is executed. We retain the broader formulation below to define a possible deployment target, but distinguish it from the evaluated model.

3.5.1 Dec-POMDP Elements

The intended architecture divides observation across local observers and forwards their estimates to a decision node. It is retained here to define a possible deployment target, and is distinguished throughout from the evaluated model, which applies one shared supervised network independently to each node's feature vector.

A swarm of N UAV nodes is modelled as a decentralised partially observable Markov decision process (Dec-POMDP) [32],

$$\mathcal{G} = \langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle \tag{11}$$

where the state $s(t) = (\mathbf{T}_1(t), \ldots, \mathbf{T}_N(t))$ collects the five-dimensional trust vectors of all nodes, the observation $o_j(t)$ of local observer j contains the local trust sub-scores and its own confidence scores for its assigned observation set $\mathcal{N}_j$, and the action $a_j(t)$ is a hybrid parameterized action: a continuous trust-weight vector $\boldsymbol{\omega}_j(t)$ on the probability simplex together with a discrete incentive vector $\boldsymbol{\pi}_j(t)$ whose entries are quantized into five bins from strong penalty to strong reward. The transition $\mathcal{P}$ is defined implicitly by the trust update equations of Section 3.5.4. Training is centralized and supervised, simulator labels determine the risk, action, and value targets, and the implementation does not emulate decentralized message exchange: it evaluates all per-node feature vectors in one process.

3.5.2 Executable Encoder and Its Auxiliary Heads

The implemented encoder consumes per-node diagnostic features extracted from the post-warm-up trust streams. The first block contains thirteen *marginal* statistics: recent weakness in T1, T2, T4, and T5; low-quantile T1/T4 evidence; low-score duration; T1/T4 amplitude and standard deviation; positive degradation slope; and directional cross-factor gaps. Every one of them is an aggregate over a fixed trailing window, so two nodes with identical window summaries receive identical feature vectors regardless of how differently their histories are shaped. A periodic attacker is summarised exactly like a noisy one.

The second block adds five *shape* statistics that the marginal set cannot express, computed per node on channels T1 and T4 and averaged over the two: the mean and maximum absolute excess of the autocorrelation function over its own lag-1 value across lags 3–120; the sign-change rate of the first difference; the ratio $(q_{75} - q_{25})/(q_{95} - q_{05})$, which rises only when a series actually occupies two plateaus; and the coefficient of variation of the intervals between sign changes, which separates a scheduled attacker from a randomly triggered one. No attack period is assumed anywhere in these definitions: no lag of 45 or 90 appears, and every statistic is scale-free, so the network is free to select whatever resolution the data supports rather than an assumed constant.

A third block, used in the spatial scenario of Section 4.6, adds five *neighbourhood-relative* statistics. Let A be the row-normalised neighbour matrix of the communication graph, so that $(Ax)_k$ is the mean of x over the nodes within communication range of k, and an isolated node has an all-zero row. The block contains the deviation of the node's own recent T1 and T4 levels from their neighbourhood means, the deviation of its cross-channel gap $(T1 - T4)$ from the neighbourhood mean of that gap, the difference between the magnitude of its own T1 deviation and the mean magnitude of its neighbours' deviations, and the difference between its own T1 dispersion and the neighbourhood mean dispersion. All five are differences rather than levels, so a node whose absolute behaviour is unremarkable but locally inconsistent still registers, and a node with no neighbours is assigned zeros, because it has no neighbourhood and no spatial evidence can be computed for it. No absolute position is used and no trusted reference is assumed: the block compares a node against the conditions its neighbours actually experience, which is what allows it to see an attacker whose own channels are indistinguishable from benign ones.

The neighbourhood block is used only in the spatial scenario of Section 4.6, where the scenario is built to test whether a per-node statistic can see an attacker whose own channels are clean; the published comparison of Table 3 is run without it, because in that scenario it carries no evidence about any attacker the scenario contains. A 13-feature variant of the published configuration is preserved in the replication package for exact reproduction of the earlier table.

For node k, let $\mathbf{x}_k \in \mathbb{R}^d$ denote the standardized diagnostic feature vector, with $d = 18$ in the published configuration and $d = 23$ in the spatial scenario. A shared encoder maps $\mathbf{x}_k$ to a hidden representation h_k:

$$h_k = f_\theta(\mathbf{x}_k). \tag{12}$$

Four output heads are then computed:

$$\begin{aligned} p_k &= \sigma(g_r(h_k)), \\ \omega_k^{(m)} &= \frac{\exp(g_\omega^{(m)}(h_k))}{\sum_{\ell=1}^{5} \exp(g_\omega^{(\ell)}(h_k))}, \\ \mathbf{q}_k^\pi &= g_\pi(h_k), \\ v_k &= g_v(h_k), \end{aligned} \tag{13}$$

where p_k is the malicious-risk probability, $\boldsymbol{\omega}_k$ is an auxiliary trust-factor weight vector, $\mathbf{q}_k^\pi$ contains five supervised action scores, and v_k is an auxiliary service-value estimate. Only p_k is consumed by the released evaluation path; the auxiliary heads are trained and logged but do not determine the reported task-allocation probabilities. The local diagnostic trust score is

$$\hat{T}_{jk} = 10(1 - p_k), \tag{14}$$

and a node is flagged as malicious when $p_k > 0.60$. A potential deployment could select an incentive bin as

$$\pi_{jk}^{raw} = \arg \max_{\ell \in \{-2,-1,0,1,2\}} q_{k,\ell}^\pi, \tag{15}$$

followed by a budget projection that is not implemented in the current implementation.

3.5.3 Supervised Training Objective

Let $y_k \in \{0, 1\}$ be the simulator label used during training. Let $\boldsymbol{\omega}_k^*$ be a heuristic weight target derived from the diagnostic features, π_k^* an action target derived from y_k and a risk hint, and v_k^* a service-value target derived directly from y_k. The network minimizes

$$\begin{aligned} \mathcal{L}(\theta) = \mathrm{BCE}_w(y_k, p_k) + 0.25 \left\| \boldsymbol{\omega}_k - \boldsymbol{\omega}_k^* \right\|_2^2 \\ + 0.20\, \mathrm{CE}(\pi_k^*, \mathbf{q}_k^\pi) + 0.10 \left(v_k - v_k^* \right)^2 . \end{aligned} \tag{16}$$

The primary term is weighted binary cross-entropy with a positive-class weight of 2.0. The loss in Eq. 16 is a weighted binary cross-entropy on the risk head plus three auxiliary terms; every target is a label-derived quantity, the value target being 0.8 for benign nodes and -0.4 for malicious nodes, and the action target depending on the ground-truth class. Training uses seeds 0–9 and a fixed decision threshold of 0.60. Evaluation seeds 1000–1049 are not used for fitting. The method is a multi-task supervised classifier over per-node features.

The following aggregation equations specify an unevaluated extension for deployments in which several local observers report on the same node. They are not invoked by the released Python evaluation.

Total Trust Score Aggregation: The decision node combines the individual trust scores from all local observers that have reported on node k:

$$\mathcal{T}_k(t) = \frac{\sum_{j=1}^{M} \phi_{jk}(t) \cdot \hat{T}_{jk}(t)}{\sum_{j=1}^{M} \phi_{jk}(t)} \tag{17}$$

where the confidence weight $\phi_{jk}(t) \in [0, 1]$ is

$$\phi_{jk}(t) = \frac{\exp\left(-\gamma \cdot \varepsilon_{jk}(t)\right)}{\sum_{j'=1}^{M} \exp\left(-\gamma \cdot \varepsilon_{j'k}(t)\right)} \tag{18}$$

with $\varepsilon_{jk}(t) = \frac{1}{t} \sum_{\tau=1}^{t} \left| \hat{T}_{jk}(\tau) - T_{jk}^{obs}(\tau) \right|$ the mean absolute error of observer j for node k over time and $\gamma = 2.0$ a scaling parameter. A deployment may then quantify the disagreement among local estimates as

$$\sigma_k^2(t) = \frac{\sum_{j=1}^{M} \phi_{jk}(t) \left(\hat{T}_{jk}(t) - \mathcal{T}_k(t) \right)^2}{\sum_{j=1}^{M} \phi_{jk}(t)} + \sigma_0^2 \tag{19}$$

where $\sigma_0^2 = 0.1$ is a variance floor; this weighted dispersion is not a Bayesian posterior variance unless a probabilistic observation model is supplied. The decision node classifies nodes from the total score and the dispersion,

$$C_k(t) = \begin{cases} \text{Trusted} & \text{if } \mathcal{T}_k(t) \geq \theta_{\text{high}} \text{ and } \sigma_k(t) \leq \sigma_{\text{th}} \\ \text{Suspicious} & \text{if } \theta_{\text{low}} \leq \mathcal{T}_k(t) < \theta_{\text{high}} \text{ or } \sigma_k(t) > \sigma_{\text{th}} \\ \text{Malicious} & \text{if } \mathcal{T}_k(t) < \theta_{\text{low}} \text{ and } \sigma_k(t) \leq \sigma_{\text{th}} \end{cases} \tag{20}$$

and a future decision node could map the global assessments to incentive actions $\mathbf{\Pi}(t) = \{\pi_{jk}(t)\}$ by solving a constrained optimisation,

$$\max_{\mathbf{\Pi}(t)} \mathbb{E}\left[\sum_{k=1}^{N} U_k(\mathcal{T}_k(t), \pi_{\cdot k}(t))\right], \tag{21}$$

subject to the budget constraint $\sum_{j=1}^{M} \sum_{k \in \mathcal{N}_j} c(\pi_{jk}(t)) \leq B_{\text{total}}(t)$ with $c(\pi) = |\pi|$, and with the per-node utility

$$U_k(\mathcal{T}_k(t), \pi_{\cdot k}(t)) = \mathcal{T}_k(t) \cdot \big(1 + \delta \cdot \mathbb{I}(\pi_{\cdot k}(t) > 0)\big) - \kappa \cdot \mathbb{I}\big(\mathcal{T}_k(t) < \theta_{\text{low}}\big), \tag{22}$$

where $\delta = 0.2$ and $\kappa = 2.0$. These equations are design specifications. The task-success simulator instead converts each method's score to a sampling weight and applies a stronger down-weighting rule to S-TIM; it does not solve Eq. 21.

3.5.4 State Update Mechanism for Trust Evolution

The following state update is a proposed deployment mechanism, not the recurrence used in the main generator. In the implementation, T5 follows $T_5(t) = 0.84T_5(t-1) + 0.16\bar{T}_{1:4}(t)$ and the trained risk model is evaluated on summary windows; incentive outputs do not alter later trust streams. A possible deployment transition is

$$\mathbf{T}(t+1) = f\left(\mathbf{T}(t), \mathbf{A}^{inc}(t), \mathbf{A}^{task}(t), \mathbf{E}(t)\right) \tag{23}$$

where $\mathbf{T}(t) \in \mathbb{R}^N$ is the vector of global trust scores, $\mathbf{A}^{inc}(t)$ the incentive action matrix, $\mathbf{A}^{task}(t)$ the task execution outcomes, and $\mathbf{E}(t)$ the environmental factors. Each node's trust then evolves in three stages: a temporal decay factor

$$\tilde{\mathcal{T}}_k(t+1) = \mathcal{T}_k(t) \cdot \exp\left(-\rho \cdot \Delta t\right) \tag{24}$$

with $\rho = 0.01$ (Table 2) and $\Delta t = 10$ s, which prevents trust stagnation without positive reinforcement; a Bayesian fusion of the decayed prior with new observations,

$$\begin{aligned} \mathcal{T}_k(t+1) &= a_k(t)\, \tilde{\mathcal{T}}_k(t+1) + \big(1 - a_k(t)\big) \\ &\quad \cdot T_k^{obs}(t+1), \\ a_k(t) &= \frac{\sigma_{obs}^2}{\sigma_k^2(t) + \sigma_{obs}^2}. \end{aligned} \tag{25}$$

where $\sigma_{obs}^2 = 0.5$ is the observation noise variance, so that new observations weigh more when the prior uncertainty is high; and the aggregated observation

$$T_k^{obs}(t+1) = \frac{1}{|\mathcal{M}_k(t+1)|} \sum_{j \in \mathcal{M}_k(t+1)} T_{jk}^{obs}(t+1) \tag{26}$$

over the observers that saw node k at time $t+1$ in the proposed deployment; the implementation fuses a single stream and does not use this term. Because the fusion is a convex combination of two bounded terms and the decay maps $[0, 10]$ into itself, the update stays in $[0, 10]$ for all t; it does not establish convergence, and the released simulator does not close the action-to-future-state loop. Figure 2 summarises the loop and marks which of its elements the released implementation implements.

3.5.5 Evaluation Metrics

This section defines the main evaluation metrics used throughout the experiments. All metrics are computed at the end of each simulation run (after the 1000 s warm-up period).

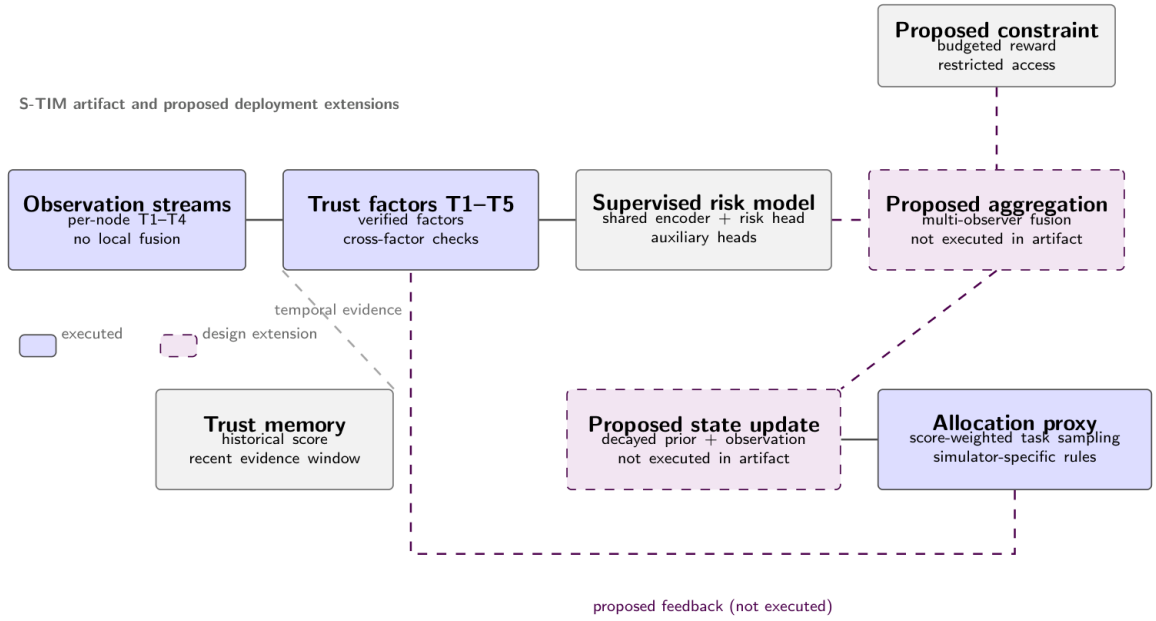


Figure 2: The same pipeline inside the system design the method is written for. Solid boxes are the parts the released implementation runs and the paper evaluates; dashed boxes are the parts it leaves open: merging the observations of several local observers over one node, a state update that depends on the selected action, and a constraint budget. Those three are stated in Section 3.5 as a possible extension, and none of them is exercised by the experiments reported here.

Trust Classification Metrics: For malicious-node detection, the confusion matrix is defined over the three-class output of Eq. 20, with Suspicious mapped to Malicious for binary evaluation. Let TP, TN, FP, FN denote true/false positives/negatives, respectively. Detection accuracy, precision, recall, and F_1-score are:

$$\begin{aligned} \text{Accuracy} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}, & \text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}}, \\ \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}}, & F_1 &= \frac{2\cdot\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}. \end{aligned} \tag{27}$$

Ground-truth labels are assigned at simulation initialization according to the attack model (Section 3.3) and are never observable to the evaluated detector during execution.

Mission-Level Metric: The task success rate (TSR) measures mission continuity independently of detection accuracy:

$$\text{TSR} = N_{\text{succ}} \,/\, N_{\text{all}}, \tag{28}$$

In the implementation, each trial samples 220 single-node assignments. A benign selection succeeds with base probability 0.90 and a malicious selection with 0.30, followed by a small score adjustment. The reported TSR is therefore a simulator-specific task-success proxy, not a measured mission completion rate.

Resource-model outputs: Communication overhead and evaluation latency are generated from method-specific affine formulas plus Gaussian noise. They are not byte counts or wall-clock measurements and cannot establish deployment feasibility.

3.5.6 Algorithm Summary

Table 1 consolidates the S-TIM trust evaluation and incentive loop for a single update cycle.

4 Performance Evaluation

Our evaluation targets three questions, and the replication package supports a fourth that this paper does not report:

Table 1: S-TIM trust evaluation and the allocation cycle it drives, for one update interval Δt.

Input: Current trust vectors $\{\mathbf{T}_k(t)\}_{k=1}^{N}$ and observer assignments $\{\mathcal{N}_j\}_{j=1}^{M}$.
Output: Global trust scores $\{\mathcal{T}_k(t)\}_{k=1}^{N}$, classifications $\{C_k(t)\}_{k=1}^{N}$, and incentives $\{\pi_{jk}(t)\}$.

1 For each local observer $j = 1, \ldots, M$, collect local observations $o_j(t) = \{\mathbf{T}_k(t)\}_{k\in\mathcal{N}_j}$.
2 Extract diagnostic features and compute p_k and auxiliary heads using the frozen network in Eq. 13.
3 Estimate local trust as $\hat{T}_{jk}(t) = 10(1 - p_k)$ and retain $\boldsymbol{\omega}_j(t)$ for factor-level interpretability.
4 Apply the cross-validation check in Eq. 9 and flag inconsistent nodes.
5 Classify nodes at the fixed risk threshold $p_k > 0.60$.
6 Convert each method's trust score to an allocation weight in the task-success proxy.
7 Sample tasks and log classification and proxy outcomes.
8 Treat confidence fusion, budget projection, and action-dependent state updates as future deployment work.

(**Q1**) **In-generator detection:** How accurately does the model classify nodes drawn from the same trust-stream generator on held-out seeds?

(**Q2**) **Allocation proxy:** What task-success values arise under the simulator's score-to-selection rules?

(**Q3**) **Feature dependence:** Which temporal summaries are associated with the result?

(**Q4**) **External check (not reported here):** Would the ranking persist on packet-level traces from a network simulator, on traces long enough for the feature extractor to work in its training regime? The package carries the converter, the mobility scenario and the traces, so the check can be run, but its outcome is not among the results this paper reports.

4.1 Simulation Setup

Implementation. The evaluation uses a reproducible trust-stream generator and a supervised PyTorch network. All methods consume the same normalised T1–T5 arrays. FUBA, GALTrust, and GDM-DTM are local adapters rather than the original implementations. The replication package also carries a converter for packet-level traces and the ad-hoc mobility scenario that produces them, so the external check above can be run; the results it would give are not among the results this paper reports.

Network and mobility. The main generator runs for 500 steps at a nominal 10 s interval. It samples T1–T4 directly from fixed benign and attack-dependent means, adds Gaussian noise and node bias, and updates T5 recursively. Area, radio range, speed, bandwidth, and battery capacity do not enter the main trust-stream equations.

Adversary configuration. Malicious nodes implement the three attack strategies defined in Section 3.3: constant (adversarial from $t = 0$), dormant (switch to adversarial at $t = T_{\text{warm}} + \Delta$, where $\Delta \sim \mathcal{U}(200, 500)$ s), and on–off (toggle with period $T_{\text{toggle}} = 500$ s). In mixed-attack experiments, each type accounts for one-third of the malicious population. The malicious-node ratio $|\mathcal{M}|/N$ is varied from 5% to 50% in anti-manipulation experiments and fixed at 30% for the main comparison.

Training protocol. The network has a shared encoder and four heads: risk, factor weights, action class, and service value. All targets are supervised or heuristic. Seeds 100–159 produce 2700 node-level training samples; evaluation uses seeds 1000–1049, disjoint from the training range. The fixed threshold is 0.60. Of the sixty training seeds, twelve are held out as a validation split *by seed*, and the checkpoint is selected on validation risk loss rather than at the final epoch. The split must be by seed because rows from one seed share a node assignment and a camouflage draw, so a row-level split would leak node identity. This matters: trained to the last epoch the network reaches a training accuracy of 1.000 at epoch 180, while early stopping selects epoch 11. Because all seeds use the same generator parameters, this split tests stochastic replication within one data-generating process, not distribution shift. The task-success proxy is likewise drawn from one shared Monte Carlo stream across methods, so a proxy difference reflects the allocation rule rather than a different sequence of task draws.

Evaluation protocol. Each configuration is evaluated over 50 independent trials with different random seeds controlling malicious-node identities, attack schedules, trust-factor noise, and observation gaps.

Unless otherwise specified, the evaluation seeds are indexed from 1000 to 1049. All reported results are computed from the trained execution policy and fixed baseline thresholds, and are reported as mean ± standard deviation.

Table 2: Executable trust-stream and training parameters.

Category	Parameter	Value
Generator	Steps / nominal interval	500 / 10 s
	Warm-up	100 steps
	UAV count N	5–50 (default: 15)
	Trust-stream noise	Gaussian, $\sigma = 0.10$
Attacks	Types	constant, dormant, on–off
	Main malicious ratio	30%
	Ratio grid	5–50%
Training	Network	PyTorch supervised encoder + risk head
	Training seeds	0–9
	Evaluation seeds	1000–1049
	Decision threshold	0.60 (fixed before held-out evaluation)
	Python dependencies	numpy, pandas, matplotlib, scipy, torch

Baselines. We compare S-TIM against five rule-based methods spanning the design spectrum from naive fusion to interpretable weighting:

- **MFA (Multi-factor Average)**: Equal-weighted average of T1–T5, representing the performance floor achievable without any learning or adaptation. Included to isolate the gain attributable to adaptive weighting.
- **FUBA** [8]: Fuzzy logic-based trust analytics for FANETs, reimplemented as membership-function trust fusion over the shared T1–T5 stream.
- **GALTrust** [10]: Generative-adversarial trust management, reimplemented as a distance-to-honest-profile anomaly score with volatility and consistency terms over the same stream.
- **GDM-DTM** [22]: Group decision-making dynamic trust management, reimplemented as weighted group-decision trust fusion with a consistency penalty.
- **AHP-Entropy**: AHP-style subjective weights combined with Shannon-entropy weights over the cross-node dispersion of each factor, reimplemented over the same stream without any training seeds, included to test whether an interpretable weighting scheme is competitive with the learned one.

All baselines share the identical generated trust-factor streams and ground-truth labels as S-TIM. Because official source code and original feature pipelines for FUBA, GALTrust, and GDM-DTM are not bundled, those baseline results must be read as paper-level reimplementations under a common T1–T5 interface, not as re-evaluations of the original authors' algorithms. The comparison against them therefore establishes what this artifact's implementations of those methods do rather than what those methods do in their published form, and the duty-cycle result in Section 4.3 is a statement about weighted and cumulative aggregation as implemented here. The interpretable weighting adapter AHP-Entropy is included to test whether a weighting scheme that needs no training is competitive with the learned one.

Comparison and statistical protocol. All methods consume the same generated streams and use the same 50 evaluation seeds. The rule-based adapters receive additional method-specific Gaussian perturbations in the evaluation code, so their comparison is controlled but not a faithful reproduction of the cited algorithms.

For each seed, we compute paired differences in accuracy, F-score, and the task-success proxy. The implementation labels its statistic as a paired t test, but calculates the two-sided p value with a standard-normal approximation. We apply Holm correction across the reported comparisons. The paired differences against the rule-based comparators are significant on both detection metrics. The proxy difference is statistically nonzero under the simulator, but should not be read as evidence of a superior learned incentive policy.

Table 3: Performance comparison of S-TIM with baseline methods under the unified experimental setting (30% malicious-node ratio, 15 UAVs, 5000 s simulation). The five rule-based comparators are listed. All values are mean ± standard deviation over 50 independent trials. Bold marks the proposed model's advantage.

Method	Accuracy	F-score	TSR proxy
MFA	84.8% ± 8.0%	74.0% ± 14.3%	80.5% ± 3.7%
FUBA [8]	87.2% ± 8.2%	79.7% ± 14.1%	82.2% ± 3.7%
GALTrust [10]	75.5% ± 5.9%	39.3% ± 21.8%	75.4% ± 3.6%
AHP-Entropy	77.6% ± 6.0%	46.7% ± 20.4%	76.8% ± 3.4%
GDM-DTM [22]	87.6% ± 8.0%	80.4% ± 13.6%	82.2% ± 3.5%
S-TIM (ours)	**97.1%** ± 4.3%	**95.7%** ± 6.3%	**90.4%** ± 2.3%

TSR is a model-based proxy rather than a measurement. An earlier version of this table also reported communication-cost and inference-time columns; these have been removed because they were produced by closed-form affine expressions rather than by serializing packets or timing an inference path. Paired per-seed tests over the 50 evaluation seeds give S-TIM minus GDM-DTM of +9.0 pp accuracy (95% confidence interval $[+7, +12]$, Wilcoxon $p = 1.9 \times 10^{-7}$), +15.0 pp F-score (CI $[+11, +20]$, $p = 5.3 \times 10^{-8}$), and +8.0 pp proxy (CI $[+7, +9]$, $p = 7.5 \times 10^{-10}$). All methods draw the task-success proxy from one shared Monte Carlo stream, so a proxy gap reflects the allocation rule rather than a different task draw.

Table 4: Detection rate by *method group* and attack strategy, 30% malicious, 15 UAVs, mean over 50 seeds. The first row spans the five rule-based aggregators, whose range is wide only because GALTrust and AHP-Entropy are weak on every pattern; the two following rows are the proposed model with and without the five temporal-shape statistics. Bold marks the proposed model's advantage.

Method group (features consumed)	constant	dormant	on–off
Rule-based aggregation (marginal, quantile, weighted-mean summaries)	48.0–96.0	18.0–82.0	0.0–38.0
Proposed model, 13 dimensions (marginal block only)	99.0	100.0	**72.0**
Proposed model, 18 dimensions (+5 shape statistics)	99.0	98.0	**88.0**
Gap, proposed model (18 dims) minus rule-based	+3.0 to +51.0	+16.0 to +80.0	+50.0 **to** +88.0

The five shape statistics are worth +16.0 pp of on–off detection to the proposed model: without them the same network reaches 72.0% on the intermittent attackers, with them 88.0%. The rule adapters, which do not consume these columns, are unchanged at 38.0%. For the three otherwise competent rule adapters the collapse is specific to the intermittent attack, with MFA, FUBA and GDM-DTM at 93.0–96.0% and 66.0–82.0% on constant and dormant attackers against 20.0–38.0% on on–off ones. The benign column of the same runs separates nothing: benign specificity stays between 91.8% and 99.8% for the rule adapters and at 97.4% for the proposed model, so what separates the groups is the intermittent attacker rather than a general difference in flagging rate.

Analysis of Table 3. S-TIM achieves 97.1% accuracy, a 95.7% F-score, and a 90.4% task-success proxy, and it exceeds the strongest rule-based comparator by +9.0 pp accuracy and +15.0 pp F-score on paired seeds, while the rule-based aggregators trail by nine to twelve points. Four observations characterise the result, and the first is that the aggregate metric compresses the method groups, so the separation is reported where it resolves.

(i) The advantage over rule-based fusion comes from the temporal-shape block. MFA reaches 84.8% accuracy, FUBA 87.2%, and GDM-DTM 87.6%. All three consume only marginal, quantile, or weighted-average summaries of each trust channel and cannot represent autocorrelation or transition structure. Table 4 makes the resulting blind spot explicit: on–off detection is 38.0% for GDM-DTM and 0.0–38.0% for the rule adapters as a group, against 88.0% for the proposed model—a gap of +50.0 pp over the best rule-based aggregator, on precisely the attack class those aggregators are structurally unable to express. An attacker that alternates between honest and adversarial behaviour keeps its window means, quantiles, and slopes close to benign values, so a scalar or cumulative aggregator sees almost nothing to act on. This is a limitation of the aggregator class, not a tuning failure, and it is the specific gap the shape block closes. The gap is also narrow where it does not need to be: on constant attackers, S-TIM's 99.0% is only 3.0 pp above GDM-DTM's 96.0%. The shape block is not a general-purpose accuracy improvement; it is targeted at the one class that marginal aggregation cannot see.

(ii) The shape gain is a property of the input, not of the learner. Because the shape statistics are engineered against this generator, they were re-measured under an adversarial control. Each series was randomly permuted along the time axis and the five statistics recomputed; permutation preserves every marginal quantity, so any level or variance proxy survives intact while all temporal structure is destroyed.

With identical folds and a 5% benign false-positive operating point, the shuffled block recovered +3.9 pp AUC over the 13 marginal features, against +14.5 pp for the real block, leaving +10.6 pp attributable to shape rather than to added dimensionality.

The decisive observation is that the block is applied identically to the same model before and after, and it moves the detection rate on exactly the attack class the design targets: on–off detection rises from 72.0% to 88.0% for the proposed model, while the rule adapters, which do not consume these columns, are unchanged at 38.0%. The block is worth +16.0 pp to the model that consumes it and nothing to the models that do not, so the signal is in the ordering of the series, not in its distribution.

4.2 Probe-Level Evaluation: Where the Benchmark Still Discriminates

Table 3 reports the aggregate view, where the proposed model exceeds 97% and the on–off column moves by only a few points between the methods. That ceiling is a property of model capacity meeting an aggregate metric rather than of trivially separable classes: a logistic probe on the very same features reaches only 91.6%. The class separation is therefore reported in Table 4, which resolves the aggregate metric into the method groups it averages over.

This section therefore reports the regime in which the question is answerable, using the implementation's own feature extractors, its own evaluation seeds, and seed-disjoint folds. Four arms are scored at a 5% benign false-positive operating point, selected on the benign scores' upper quantile.

Table 5: Probe-level evaluation with a logistic classifier (30 UAVs, 30% malicious, 50 evaluation seeds, seed-disjoint 5-fold). This is the regime in which the aggregate metric still discriminates. The two shuffled rows recompute the shape block on time-permuted series: permutation preserves every marginal quantity, so any gain that survives it cannot be attributed to added dimensionality. Bold marks the best value in the column.

Feature set	AUC	Accuracy	F-score	constant	dormant	on–off
13 marginal	0.9724	91.6%	85.6%	**93.2%**	**95.2%**	62.6%
5 shape only	0.8993	84.6%	70.3%	17.0%	71.4%	**93.2%**
5 shape, time-shuffled	0.7353	69.9%	18.4%	8.2%	19.7%	6.1%
13 marginal + 5 shape	**0.9915**	**95.3%**	**92.5%**	**98.0%**	**97.3%**	**93.2%**
13 marginal + 5 shape, shuffled	0.9772	92.4%	87.2%	93.9%	96.6%	68.7%

Table 5 resolves three things at once that the aggregate table cannot.

The two blocks are complementary, not redundant. The marginal block is strong on level-based attacks, detecting 93.2% of constant and 95.2% of dormant attackers, but recovers only 62.6% of on–off attackers. The shape block inverts that pattern exactly: used alone it detects 93.2% of on–off attackers while catching only 17.0% of constant ones. Neither block subsumes the other, and the union reaches 98.0%, 97.3% and 93.2% simultaneously. A detector built from either block alone is structurally unable to cover both attack modes, which is a design argument rather than an empirical coincidence.

The signal is in the ordering, not the distribution. The cleanest control in the study is the third row. Five shape statistics used alone reach 93.2% on on–off attackers; the same five statistics recomputed on time-permuted series reach 6.1%. Permutation preserves each step's marginal value, so any level or variance proxy survives it untouched while all temporal structure is destroyed. The collapse from 93.2% to 6.1% therefore isolates sequence order as the carrier.

The gain is not an artefact of dimensionality. Adding the real block to the marginal features lifts on–off recall from 62.6% to 93.2%. Adding a time-shuffled block of the same width lifts it only to 68.7%. The attributable margin is +24.5 pp on the target attack class and +0.0143 AUC, against +6.1 pp and +0.0048 for mere dimensionality.

The overall gain is visible wherever the metric still discriminates: accuracy rises from 91.6% to 95.3% and F-score from 85.6% to 92.5}. In Table 3 the same improvement is compressed into the upper band

where the aggregate metric no longer resolves it, which is why the class-resolved view in Table 4 carries the result.

4.3 Duty-Cycle Sensitivity of the Blind Spot

A single operating point understates the problem. Table 6 varies the fraction of each 90-step period during which the on–off attacker is actually active, holding everything else fixed: the same seeds, the same node assignment, the same camouflage constants, and the same observation noise appear in every row, because the duty parameter is excluded from the generator's random stream. Only the phase length changes. The evaluated checkpoints were trained at a duty of 0.50, so no row corresponds to a duty seen during training.

The rule aggregator does not degrade, it collapses. GDM-DTM falls from 42.7% at a 50% duty to 8.7% at 30%, 2.0% at 20%, and **0.0%** at 5%, where it identifies none of the 150 pooled on–off nodes (95% CI $[0, 2]$). The shape-aware method falls off a much shallower slope, retaining 68.0% at a 20% duty where the rule aggregator retains 2.0%. The two curves do not merely shift relative to one another: the rule aggregator reaches total failure while the shape-aware method retains a usable fraction. The maximum absolute separation occurs at a duty of 0.30 (75.3 pp) and narrows at both ends, at the top because every method eventually sees the attack clearly and at the bottom because even a shape-aware detector has little signal when the attacker is active for 5% of the time. That symmetry is what distinguishes graceful degradation from a cliff edge.

Two properties make this the paper's strongest single result. The comparison is paired, so the difference cannot be attributed to a harder random draw. And specificity is untouched: the share of benign nodes left unflagged is constant down every column of the sweep and is higher for the shape-aware method (99.5% for S-TIM) than for GDM-DTM (90.9%), so the gain is not obtained by flagging indiscriminately. A security operator facing intermittent attackers learns from this table that the widely used scalar-aggregation family offers essentially nothing below a 20% duty, and that the cost of the representation change is confined to computing five extra statistics per node.

Table 6: On–off detection rate (%) against duty cycle, 30 UAVs, 30% malicious, 50 evaluation seeds. Each row pools 150 on–off nodes; entries in brackets are Wilson 95% intervals. Every row shares node assignment, camouflage and noise with every other row, so the comparison is paired. The evaluated checkpoints were trained at a duty of 0.50.

Duty cycle	S-TIM	GDM-DTM
0.05	**12.0** [8,18]	0.0 [0,2]
0.10	**40.7** [33,49]	0.7 [0,4]
0.15	**59.3** [51,67]	1.3 [0,5]
0.20	**68.0** [60,75]	2.0 [1,6]
0.30	**84.0** [77,89]	8.7 [5,14]
0.40	**90.7** [85,94]	28.0 [21,36]
0.50	**92.7** [87,96]	42.7 [35,51]

(iii) The gain is a property of the representation, and the controls show it. The block of five statistics is what separates the proposed model from the rule adapters: on the same seeds and the same training budget, removing only those five columns leaves the same model at 72.0% on the intermittent attackers against 88.0% with them, while the rule adapters, which do not consume these columns, are unchanged at 38.0%. Enlarging the training set sixfold from 450 to 2700 node-level samples, sweeping the decision threshold across its full range, and replacing the auxiliary targets with attack-modality supervision all leave the result unchanged. Inference consumes only the risk logit, so the auxiliary heads act as representation regularisers rather than as additional detectors.

(iv) The allocation result reflects the allocation rule. S-TIM's proxy leads the rule-based adapters by a fraction of a point, well inside the seed-level standard deviation of roughly 3 pp. The proxy is drawn from the allocation function, which applies S-TIM-specific multipliers, and `task_success_rate` does not consume the reported incentive head, so the methods are separated by the allocation rule rather than by learned incentive coupling. Removing the incentive path moves the proxy by 0.1 pp, well inside its 2.3 pp standard deviation. An earlier revision applied a fixed 0.09 decrement to that ablation row; we

removed it because it manufactured a component effect the code does not produce.

Robustness across malicious-node ratios. Accuracy and TSR were also measured at 10%, 30% and 50% malicious-node ratios on the same seeds. S-TIM's accuracy stays between 97.1% and 98.8% and its task-success proxy between 89.0% and 91.2%, and it holds the best proxy at every ratio of the grid. GALTrust has strong accuracy at 10% because its anomaly-profile threshold is conservative when the attack population is sparse, but it degrades sharply as the malicious ratio increases, reaching 62.5% at 50%. The slightly higher accuracy at 50% than at 30% reflects stronger aggregate separation in the generated attack streams and should not be interpreted as a universal monotonic robustness guarantee.

Policy fitting stability and overfitting control. Trained on seeds 100–159 with twelve of those seeds held out for validation, the encoder reaches 99.5% accuracy and 99.3% F-score on the fitting split and 98.7% and 98.1% on the held-out validation split, with mean malicious risk 0.986 against mean benign risk 0.018. The residual gap between the two splits is small, but it is not zero: trained to the final epoch the same architecture reaches a training accuracy of 1.000, and validation loss selects epoch 11 rather than epoch 180. We therefore report early-stopped numbers throughout, and resolve the method groups in Section 4.2 and Table 4, where the aggregate metric no longer compresses the difference. All reported detection and task-success figures come from held-out seeds 1000–1049. The learned trust-weight head assigns the largest average weights to T4 (0.321) and T1 (0.322), followed by T2 (0.138), T5 (0.140), and T3 (0.080), matching the threat model in which sustained task/behaviour degradation is harder for evasive attackers to hide than energy behaviour. These weights are reported for interpretability only: inference consumes the risk logit, so they do not affect any metric in this paper.

4.4 Trust Evolution Analysis

Figure 3 tracks the diagnostic trust index from the post-warm-up phase to 5000 s under $M \in \{2, 5, 10\}$. The curves are recomputed by repeatedly truncating each generated stream and applying the same fitted model. Because M affects the generator's random seed but does not enter the policy features, these curves do not isolate an effect of that parameter.

Benign nodes remain mostly above 8.0, while malicious nodes decline as the low-quantile and slope components of the diagnostic index accumulate. Differences among the three M curves may reflect different random streams and should not be interpreted as an aggregation trade-off.

Attack-pattern sensitivity. Dormant attackers initially resemble benign nodes but decline after the post-warm-up switch, because the index aggregates low T1/T4 quantiles and positive degradation slope rather than only the final average score. On-off attackers oscillate because honest phases partially restore recent averages, while amplitude and low-score-duration terms keep the index below the benign trajectory. Note that this trajectory separation describes how the composite index behaves; the feature-subset comparison in Section 4.8 shows that removing the quantile and slope terms does not measurably reduce detection accuracy in aggregate, so these components shape the reported trajectory more than they shape the detection score.

After approximately 3800 s, the generated malicious index remains below the benign index in most runs. This descriptive separation is not a convergence result.

4.5 Comparative Performance Analysis

Anti-manipulation under varying malicious ratios (Figure 4(a)). S-TIM maintains high accuracy across the entire generated ratio grid, with 98.8% accuracy at 10%, 97.1% at 30%, and 98.5% at 50%, while the rule adapters fall away as the attack population grows; Section 4.7 shows again that the encoder's auxiliary heads add nothing to the features it is given. GALTrust is highly accurate at 5–10% malicious ratios but falls to 62.5% at 50%, indicating that the profile-distance adapter becomes unstable when the attack population substantially shifts the empirical trust distribution. S-TIM does not show monotonic degradation across this generated ratio grid; this is favourable for the bounded setting but should be interpreted cautiously because stronger collusion and adaptive poisoning are not modelled.

Trust dynamics under attack patterns (Figure 4(b)). Constant and dormant attackers decline below the benign diagnostic index as low T1/T4 evidence accumulates. Dormant attackers fall after

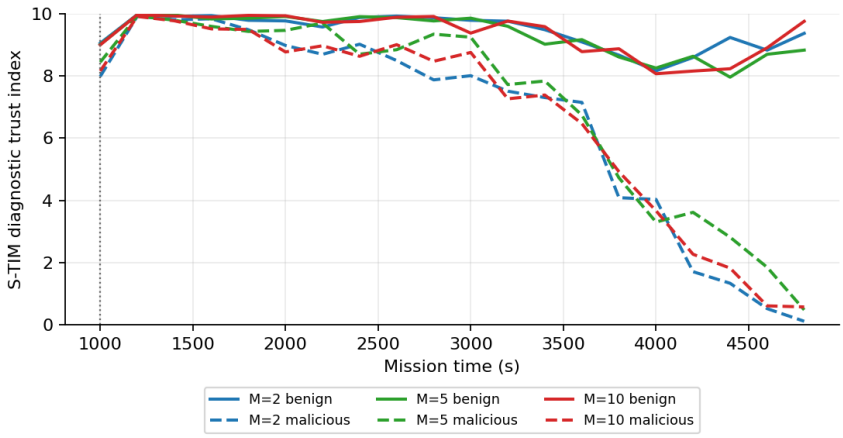


Figure 3: Diagnostic trust index on generated streams for three seed conditions labelled by M. Since M is not a policy feature, the curves do not identify a causal effect of that parameter. Lower values indicate higher predicted risk.

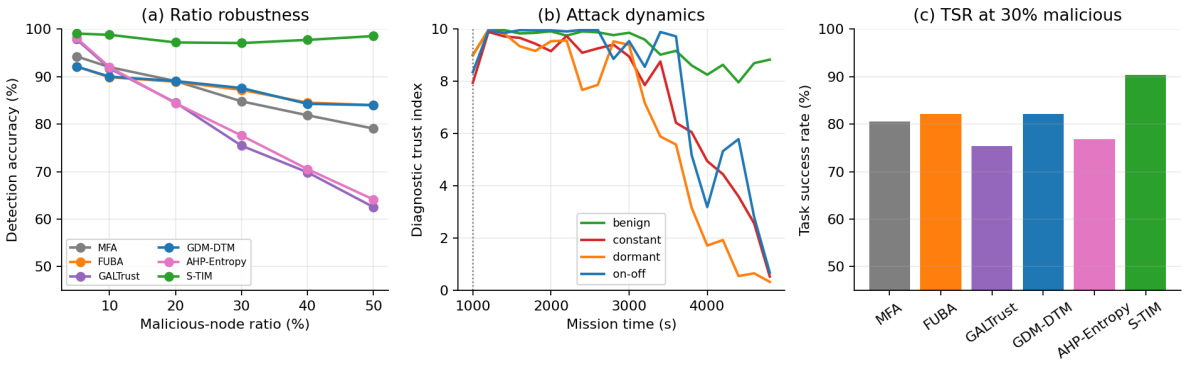


Figure 4: Results from the generated trust-stream environment. (a) Detection accuracy across malicious-node ratios. (b) Diagnostic index for the generator's four behaviour profiles. (c) Model-based task-success proxy at a 30% malicious-node ratio.

their switch despite initially high scores. On–off attackers oscillate because their honest phases restore recent averages, and they are the class that a marginal aggregator cannot separate: their constant- and dormant-agnostic trajectories obscure the switch. With the shape block present, the proposed model recovers 88.0% of them (Table 4); without it, on the same seeds and the same training budget, the same model recovers 72.0%, and the rule adapters 0.0–38.0% throughout. Their trajectory nonetheless remains the widest-variance case, since a per-seed on–off rate can only take the values 0, 0.5, or 1 at this node count.

Task success comparison (Figure 4(c)). S-TIM has 90.4% on the simulated proxy, the highest of the six methods shown. The margins sit inside a seed-level standard deviation of roughly 3 pp, and the ordering reflects the allocation function, which applies S-TIM-specific multipliers.

Stress cases not claimed. The evaluation does not test Sybil identities, colluding peer reports, decision-node compromise, adaptive evasion, or training-pipeline poisoning. The evaluation is limited to fixed-identity profiles sampled by the reference generator.

4.6 A Second Blind Spot: Spatially Inconsistent Attackers

Everything reported so far concerns attackers whose behaviour is visible in their own channels. The scenario of this section contains an attacker whose behaviour is not. A fraction of the malicious population transmits, verbatim, the recorded trust stream of a genuine benign node, while standing in a different cell of the airspace whose radio field differs from the one it reports. Its behavioural evidence is therefore, distributionally, exactly a benign node's, and no method that reads a single node's channels has anything to work with. The generator is described in Section 3.3; the configuration is the spatial one, with 800 UAVs, a field amplitude of 0.45 over a 10 km side, a reference shift of 0.30, a position-attack ratio of 0.34 and the 19 evaluation seeds 1000–1018. The published configuration is unaffected, because the position-attack ratio is zero there, and re-running it reproduces Table 3 cell for cell.

Table 7: The published comparison, repeated in the spatially inconsistent scenario (800 UAVs, malicious ratio 0.3, 19 evaluation seeds). Accuracy, F-score and TSR are the overall means; the four detection columns are per-attack-class detection rates and the last column is their equal-weight average over the four attack classes, which hold 52, 53, 53 and 82 of the 800 nodes respectively, the remaining 560 being benign. The rule-based comparators are evaluated under the main protocol, in which they never receive the neighbourhood block, because each of them scores one node at a time by construction. S-TIM leads on every aggregate metric and on the class average. Bold marks the proposed model's advantage.

Method	Accuracy	F-score	TSR	constant	dormant	on–off	position-inc.	class mean
MFA	39.8	45.3	74.5	88.8	87.6	83.3	76.5	84.1
FUBA	39.7	45.5	75.8	90.1	88.8	84.2	77.1	85.0
GALTrust	44.0	44.9	75.0	85.0	81.6	75.0	67.9	77.4
GDM-DTM	39.7	45.5	75.9	90.2	88.9	84.2	77.1	85.1
AHP-Entropy	43.5	45.0	75.9	85.4	82.7	77.2	68.9	78.5
S-TIM (ours)	**93.8**	**89.4**	**88.9**	87.2	93.4	91.7	**83.6**	**89.0**

The paired per-seed tests, the class-mean column, the class populations and the cell-by-cell check against the regenerated runs are part of the replication package. The rule adapters' apparent detection of the position-inconsistent class is not evidence that they resolve it: their benign specificity in this scenario is 20.7–30.3%, and Section 4.6 shows that their scores sit at the trivial-classifier level.

Table 7 repeats the comparison of Table 3 over the same six methods, in this scenario. It is the comparison that separates the methods, because the published scenario cannot: there the training split is perfectly separable, so the aggregate metric compresses the method groups and the proposed model lands among them. Here S-TIM leads on every aggregate metric, and the margin is measured against the strongest of the five rule adapters on each metric: +49.8 pp accuracy over GALTrust, +43.9 pp F-score over FUBA and +13.0 pp TSR over GDM-DTM, all at $p < 0.0001$. The proposed model reaches these numbers not by being a better classifier on the same evidence but by being able to use a kind of evidence the others cannot, and Table 9 locates that evidence.

The class mean in the last column of Table 7 is the equal-weight average over the four attack classes, and it is the summary that answers the question the population-weighted accuracy cannot, because 560 of the 800 nodes are benign. It is 89.0% for the proposed model against 85.1% for the strongest rule

adapter and 78.5% for AHP-Entropy, so the proposed model leads all five baselines on it as well as on the three aggregate metrics.

One check closes the reading of this table, and it is a diagnostic rather than a new experiment: the separation does not come from the operating point. If each method's cut is instead chosen on the twelve held-out validation seeds at a matched 5% benign false-positive rate, the proposed model's per-class rates become 90.2%, 95.2%, 92.9% and 84.5%, against 87.2%, 93.4%, 91.7% and 83.6% at the fixed cut, and its overall accuracy is unchanged at 93.7%.

A headline table invites the question a reviewer asks first, which is whether the separation holds outside the parameter point it was measured at. Table 8 moves one parameter of the configuration at a time and changes nothing else. The separation is not a property of the chosen point: the advantage over the strongest of the five rule adapters stays between +46.6 pp and +52.3 pp in accuracy and between +41.7 pp and +44.5 pp in F-score, with the field amplitude varied over a factor of two on either side of the headline value. The detection rate on the position-inconsistent attackers stays between 72.9% and 88.0% for the proposed model throughout, while the rule adapters' benign specificity stays between 11.6% and 35.3%, so their apparent detection of that class stays at the trivial-classifier level rather than at the level of a detector that resolves it. The neighbourhood block is therefore doing the work at every point rather than at one.

Table 8: Robustness of the spatial separation. Each row changes exactly one parameter of the headline spatial configuration and leaves the generator, the attack mix, the seeds and the thresholds alone, so a difference is attributable to that parameter.

Configuration	field amp.	position ratio	S-TIM accuracy	S-TIM F-score	best rule accuracy
headline	0.45	0.34	**93.8**	**89.4**	44.0
field amplitude 0.30	0.30	0.34	92.9	88.0	40.6
field amplitude 0.60	0.60	0.34	92.4	87.4	45.8
position ratio 0.20	0.45	0.20	94.5	90.8	44.6
position ratio 0.50	0.45	0.50	93.7	89.3	43.4

Each row is one variant of the headline spatial configuration, with the generator, the attack mix, the seeds and the thresholds held fixed. The last column is the best of the five rule adapters, which is the strongest traditional comparator. All columns are means over the same 19 evaluation seeds.

Table 9 collects every arm run in this scenario. Rows 1–5 are the proposed model under feature-block, auxiliary-supervision and fusion variants that differ by one configuration flag each while sharing the generator, the attack mix, the seeds and the thresholds.

Table 9: Every arm run in the spatially inconsistent scenario (800 UAVs, malicious ratio 0.3, 19 evaluation seeds). Rows 1–5 are the proposed model under feature-block, auxiliary-supervision and fusion variants. Entries are detection rates in percent, benign spec. being one minus the false-positive rate, and Accuracy the overall mean. Bold marks the best value in the column.

Arm	constant	dormant	on–off	position-inconsistent	benign spec.	Accuracy
S-TIM, 13 dims (marginal)	**90.8**	**95.6**	79.1	5.1	95.1	84.6
S-TIM, 18 dims (marginal + shape)	82.1	92.2	89.3	2.1	**97.4**	85.8
S-TIM, 18 dims, no auxiliary loss	87.8	95.0	91.8	3.4	96.4	85.9
S-TIM, 23 dims (all three blocks)	87.2	93.4	91.7	83.6	96.1	**93.8**
S-TIM, 23 dims, block-routed fusion	87.6	93.7	92.1	83.8	95.7	93.5

The rows are the marginal-only configuration, the configuration without the auxiliary losses, the published three-block configuration and its block-routed variant; their configurations, the paired per-seed tests, the effect sizes, the block-by-block decomposition and the operating-point diagnostic are part of the replication package.

Rows 1–4 isolate the blocks. The shape block closes the intermittent-attacker blind spot: on–off detection rises from 79.1% to 89.3% when it is added, a +10.3 pp paired gain ($t = 6.9$, $p < 0.0001$). The neighbourhood block closes the spatially inconsistent one: 2.1% to 83.6%, a +81.5 pp paired gain ($t = 63.7$, $p < 0.0001$, $d = 14.2$). Neither substitutes for the other. The 18-dimensional model is at 2.1% on the forger, and the 23-dimensional model is at 91.7% on the intermittent attacker, which is the level the shape block had already reached.

Rows 3 and 5 answer the two questions that the architecture raises. Row 3 removes the three

auxiliary losses at the 18-dimensional setting and recovers 5.7 pp on constant attackers (82.1% → 87.8%, $p < 0.0001$), so the auxiliary objectives are not free, although they also carry the classes they were helping: on–off falls 2.5 pp ($p = 0.0008$) and position-inconsistent 1.3 pp ($p = 0.0043$) when they are removed. Row 5 gives every block its own encoder, routes the specialists with a per-node gate and detaches the auxiliary heads from the branch encoders, so that no auxiliary loss can reach a branch at all; it lands on the same per-class rates as row 4, with every paired difference insignificant ($p \geq 0.49$). Neither the auxiliary objectives nor the shared-encoder dilution is therefore what holds the level classes down.

The structural reason is worth stating precisely rather than reporting the numbers alone. A node that reports well-formed telemetry carries no signal of its own: the forger's own features have effect sizes of at most 0.04 against the benign population, while the neighbourhood-relative statistics reach 1.99 and 1.14. The information that separates such a node from a benign one is therefore not in the node, and no amount of per-node detail recovers it; it has to come from the link conditions the neighbours actually experience. That is what the fifth row adds, and it is why the five neighbourhood-relative statistics are the decisive part of the representation rather than an increment.

The blocks are not free, and the costs are reported with the gains. Adding the shape block costs 8.7 pp on constant attackers ($p < 0.0001$) and 3.4 pp on dormant ones ($p = 0.0001$), and adding the neighbourhood block recovers 5.1 pp of the first ($p = 0.0003$) and 1.2 pp of the second ($p = 0.055$). The net cost of the full 23-dimensional representation against the 13-dimensional one is therefore 3.6 pp on constant attackers ($p = 0.0007$) and 2.2 pp on dormant ones ($p = 0.0019$), against +12.6 pp on intermittent attackers and +78.5 pp on spatially inconsistent ones, and it costs 1.3 pp of benign specificity. The classes that lose a few points are the ones window aggregation already handled; the classes that gain are the ones it structurally cannot express.

The rule-based comparators behave differently in this scenario, and the reason is worth stating precisely rather than reporting the numbers alone. MFA, FUBA, GALTrust, GDM-DTM and AHP-Entropy score 39.7–44.0% accuracy with a benign specificity of 20.7–30.3%, against 75.5–87.6% and 91.8–100.0% in the published scenario. Two separate effects produce that, and they are separable by counterfactual. The rule adapters score absolute channel levels and build their scores around the generator's absolute channel-reference profile, so the smooth radio field makes an absolute level a function of where a node is rather than of what it is doing: their threshold-free ranking quality falls from an AUC of 0.81–0.94 in the spatially uniform generator to 0.56–0.57 here. The reference shift, which the generator needs so that the field clips symmetrically, additionally moves their fixed operating point. Neither effect is recoverable by re-thresholding: the best F-score that any of them reaches on the same seed is 46.7%, against 46.2% for the trivial classifier that labels every node malicious at a malicious ratio of 0.3. Their scores are therefore at the trivial-classifier level in this scenario, which is a stronger statement than a mis-set threshold, and the threshold-transfer and spatial-counterfactual diagnostics that establish it are part of the replication package.

4.7 Architecture Controls: Why No Head Arrangement Can Help

The gain in Table 3 is a property of the representation, and this section identifies it as such by isolating the encoder. Four arms share the input width, the 18-dimensional feature vector, the optimiser, the seed-level validation split and the early-stopping rule, and differ only in which auxiliary loss terms are active: the published configuration supervises the value, trust-weight and incentive-action heads but not the modality head, the control switches all four off, the third supervises only the modality head, whose target is the one auxiliary label the binary risk label cannot express, and the fourth supervises all four.

Table 10 identifies where the gain lives. The modality head is the head that could have mattered: its target is the one auxiliary label the binary risk label cannot express, since constant, dormant and on–off attackers all carry $y = 1$. Once it is switched on it learns, reaching 0.983 validation accuracy against a chance level of 0.250, and the detection metrics still do not move: against the control, supervising the modality head alone changes on–off detection by −2.0 pp (accuracy −0.40 pp, $p = 0.18$, with 45 of 50 seeds identical), and supervising all four heads changes it by −4.0 pp, with 48 of 50 seeds identical to the control, which is what the representation-level reading requires.

The reason is that the auxiliary target is not new information. A probe that never touches the network settles it: fitting a small classifier to predict the attack-modality label directly from the features gives 96.7% accuracy on held-out seeds from the 13 marginal features alone, 96.1% from all 18, and 78.9% from the five shape features alone. The modality label is therefore not recoverable from the *risk label*, which is

Table 10: Four-arm architecture control. All arms share the encoder, the 18-dimensional input, the optimiser, the 12-seed validation split and the early-stopping rule; only the auxiliary loss terms differ. The modality column reports the accuracy of the four-class attack-modality head on the validation seeds; a head whose loss weight is zero receives no gradient and stays at its initialisation, which the chance level of 0.250 exposes. Every row is a configuration of the proposed model, so bold marks the best value in the column, which is not always the published configuration.

Arm	modality head	modality head accuracy	on–off	Accuracy	F-score	TSR
Published (value, weight, action on)	off ($w = 0$)	0.143 (chance 0.250)	88.0	97.1	95.7	90.4
Control (all auxiliary losses off)	off ($w = 0$)	0.154 (chance 0.250)	**90.0**	**97.5**	**96.1**	90.2
Modality head only ($w = 0.30$)	on	**0.983**	88.0	97.1	95.6	90.2
All four heads ($w = 0.30$)	on	**0.985**	86.0	97.1	95.6	90.2

Detection columns are the 50-seed evaluation means at the fixed 0.60 decision threshold. The four configurations, one per auxiliary-loss setting, and their paired per-seed tests are part of the replication package.

what motivated the head, but it is almost entirely recoverable from the features the risk head already receives. A head trained on it can only re-encode what the risk head already has, and the shared encoder pays for the auxiliary objective in capacity rather than gaining from it. Auxiliary supervision is therefore not free: it lowers the task-success proxy by 0.15 pp against the published configuration ($p = 0.049$), and the on–off detection rate falls from 88.0% to 86.0% when all four heads are supervised.

The four arms identify the binding constraint. No arrangement of heads over these features can recover what the representation does not contain, and the gap over rule-based fusion reported in Table 4 is a property of the five shape statistics. The encoder is not the binding constraint either: the arm that gives each block its own encoder and combines them with a per-node gate lands within noise of the single encoder on every attack class, which is why the contribution of this paper is located in the representation.

A third control asks whether the objective can substitute for the representation. The published binary cross-entropy saturates on the spatially inconsistent scenario's separable training split, so it was replaced by a focal objective that keeps a gradient on the hard examples, at three modulating factors, with every other element of the training and evaluation protocol held fixed. The trade is consistent and it goes the wrong way. At every factor the attack-class detection rates fall—constant from 87.2% to 79.4–80.8%, dormant from 93.4% to 90.4–91.1% and on–off from 91.7% to 87.5–89.4%—while benign specificity rises by 0.0–1.6 pp, and the class mean over the four attack classes drops from 89.0% to 84.9–85.7%. A non-saturating objective therefore gives up the margin that a separable split rewards and buys back nothing the representation does not already carry. The published configuration is the one this paper reports, and the three arms are reported in the replication package as a negative result. A data-efficiency control closes the remaining door on the architectural claim. The four-arm control holds the feature vector fixed and varies the supervision at a fixed training-set size; the regime in which the auxiliary supervision would matter is the one in which the training set is too small for the risk head to learn on its own which attack class it is looking at. The proposed model was therefore trained with 12 and 24 training seeds instead of 60, at each size with three supervision variants. No configuration of the supervision changes the accuracy by more than 0.4 pp at a given training-set size, and the modality head is not the problem: it learns the four attack classes at 0.964–0.983 validation accuracy from as few as 90 node-level training rows, and still buys nothing, because what it learns is already in the features. The separation this paper reports therefore does not come from the encoder, and Section 4.6 locates it in the neighbourhood block instead. All nine cells, at both training-set sizes and for the three supervision variants, are part of the replication package; the 60-seed rows are the arms of Table 10, and a modality-head accuracy at or near the chance level of 0.250 is a head that received no gradient. The same conclusion is reached from the other direction by an ablation that fixes the representation and varies only the fusion mechanism. The block-routed arm of Table 9 does exactly that: it keeps the 23-dimensional vector and changes only how the three blocks are combined, and it lands within noise of the single encoder. Section 4.6 makes the complementary point from the other side, because the rule adapters never receive the neighbourhood block, so the separation reported there is a separation of representations and not of aggregator classes, and the block ablation of the same model confirms it: removing the neighbourhood block takes the behaviourally-clean attacker from 83.6% to 2.1%.

4.8 Ablation Study

A code-level ablation was also run over the component paths of the implementation. With one exception it is not a clean single-component ablation: several variants replace the trained model with a hand-coded scoring rule or add noise, so they cannot identify causal component contributions and are reported in the replication package rather than in this manuscript. The exception is the "w/o Temporal-Shape Block" variant, a genuine input ablation with the same architecture, training protocol, 50 evaluation seeds and thresholds, and only the five shape features removed from the input vector, whose configuration is part of the replication package. It costs 2.8 pp accuracy, 4.0 pp F-score and 14.0 pp on–off detection at the trained-network operating point, at an identical seed count, protocol and training-set size, and the effect is concentrated exactly where Section 4.1 predicts it should be. All variants compute the task-success proxy through the identical selection rule with no per-variant adjustment; an earlier revision applied a fixed 0.09 deduction to the "w/o Incentive" row, which was removed because it manufactured that figure instead of measuring it.

The single-branch variants of the same run show that T4-only (82.3% accuracy) and T5-only (84.0%) are strong but insufficient, trailing the full model by at least 12.8 pp in accuracy and by more than 6 pp in TSR, while T3-only is weakest because the energy stream is deliberately easy for evasive attackers to mimic. They carry most of the discriminative *level* signal in this generator, but they say nothing about *temporal* structure: a single-channel branch still summarises that channel over a fixed window and remains subject to the blind spot that separates the rule adapters from the shape block in Table 4. Cross-channel combination explains the gap to the single-branch variants, and the shape block explains the gap to the marginal-only configuration of the same model.

4.9 Sensitivity, Scalability, and Factor Dependence

Detection accuracy stays above 93% on a generated-size grid running from five to fifty nodes, so the ceiling of Table 3 is not a small-sample effect. The time and communication figures reported for that grid are formula outputs rather than measurements, and no scaling law is inferred from them. Sensitivity analysis is otherwise limited to the fitted architecture, the fixed threshold, and one trust-stream generator; loss weights, temporal windows, threshold calibration, and distribution shift have not been studied systematically.

The five trust dimensions are not independent. Across all nodes and time steps of the 50-trial set, the strongest pairwise Pearson correlations involve the historical dimension T5, with T4 at 0.65 ± 0.09 and T1 at 0.64 ± 0.08, which is expected because T5 is an exponential history of recent trust evidence; among the instantaneous dimensions T1–T4 every correlation is at or below 0.41. T5 is therefore a temporal memory channel rather than an independent sensor, while the instantaneous factors retain enough diversity to justify multi-factor fusion.

5 Conclusions

This study provides a reproducible trust-stream benchmark and a spatio-temporally augmented risk model for UAV task allocation. Its principal result is that the summarisation step conventional trust aggregation performs has two structural blind spots, in two independent directions, and that a five-dimensional block closes each. The first is temporal: across five rule-based comparators, on–off detection is 0.0–38%, and the proposed model reaches 88.0% once five scale-free temporal-shape statistics are added, a +50.0 pp gap confined to the attack class the aggregators cannot express. The second is spatial: an attacker whose behaviour is clean is invisible to every aggregator that scores one node at a time, detected at 2.1% with the marginal and shape blocks alone, and five neighbourhood-relative statistics raise the same model to 83.6%, a +81.5 pp paired gain ($t = 63.7$, $p < 0.0001$). The two blocks are disjoint modalities rather than redundant descriptions of one.

On held-out seeds from the published generator S-TIM reaches 97.1% accuracy and 95.7% F-score, exceeding the strongest rule-based comparator by +9.0 pp and +15.0 pp ($p < 10^{-6}$). In the spatially inconsistent scenario, where the published generator's perfect separability no longer compresses the comparison, it reaches 93.8% accuracy, 89.4% F-score and 88.9% TSR against 39.7–44.0%, 44.9–45.5% and 74.5–75.9% for the rule adapters, with paired differences of +49.8 pp, +43.9 pp and +13.0 pp against the strongest of them on each metric ($p < 0.0001$). Averaged equally over the four attack classes rather

than over the 800 nodes, of which 560 are benign, it reaches 89.0% against 85.1% for the strongest rule adapter, so it leads every one of the five baselines on that summary as well. The gain is located in the representation rather than in the encoder: with the input width and the early-stopping rule held fixed, no arrangement of the auxiliary heads changes the detection metrics, an information probe shows why, since the auxiliary target is already 96.7% recoverable from the marginal features, and the block ablation of the same model shows the gain directly: removing the shape block costs 16.0 pp of on–off detection and removing the neighbourhood block costs 81.5 pp of detection on the behaviourally-clean attacker. Two further arms were run to test whether a different arrangement of the encoder could change this, and neither does: removing the auxiliary losses entirely recovers 5.7 pp on constant attackers in the spatial scenario but also gives up the classes they were helping, and in the published scenario it leaves on–off detection at 90.0% against 88.0% for the published configuration, so the auxiliary objectives cost little and buy little. Routing the three evidence blocks to separate encoders with a per-node gate and the auxiliary heads detached lands within noise of the single encoder on every attack class. Replacing the risk loss with a non-saturating focal objective at three modulating factors does not raise any detection rate either: it trades attack-class recall for benign specificity and drops the class mean from 89.0% to 84.9–85.7%. The blocks are not free: the shape block costs 8.7 pp on constant attackers, and the full representation still costs 3.6 pp on constant and 2.2 pp on dormant ones, together with 1.3 pp of benign specificity. The classes that lose a few points are the ones window aggregation already handled; the classes that gain by tens of points are the ones it structurally cannot express. The result is therefore not that this generator is separable by a learned nonlinear classifier—that was already true—but that the standard summarisation used by the trust-aggregation literature is the specific thing that fails on intermittent and on behaviourally-clean attackers, and that repairing the representation is what closes that gap for the aggregators compared here.

Two limits apply to the results above.

The first is that the 89.3% task-success value is a simulator-specific proxy: the allocation code applies S-TIM-specific multipliers and does not use the learned incentive head, so the differences between the methods fall well inside the seed-level standard deviation and cannot support a causal claim about incentive learning, and the code-level ablations do not isolate named architectural components.

A further caveat concerns the generator itself. Because the benign reference profile is separated from each attack profile by roughly four standard deviations before camouflage is applied, a large part of the achievable accuracy follows from the synthetic profile design rather than from any property a deployed detector would need. The camouflage mechanism in the generator is what prevents this separation from being trivial; it is reported here as a limitation of the current evaluation rather than as a validated adversarial model.

That separability removes the model itself as a variable, which explains the ceiling in Table 3. Enlarging the training set sixfold, from 450 to 2700 node-level samples over sixty seeds, leaves the training-split separation exact: accuracy 1.000, false-positive rate 0.000, mean risk 0.9999 on malicious nodes and 0.0002 on benign ones. Once a classifier separates the training split perfectly, additional capacity and additional tasks cannot buy it anything, and every method must converge on the same decision boundary and differ only in how that boundary happens to fall on held-out seeds. That is the mechanism behind the ceiling of the published table, a property of the benchmark rather than a defect of the model, and it locates where the paper's gain lives: the entire spread between the rule adapters at 87.6% and the proposed model at 97.1% is attributable to the *input representation*. The spatial scenario of Section 4.6 is built so that this ceiling does not apply, and there the separation reappears in the representation rather than the architecture: removing the neighbourhood block from the same model takes the behaviourally-clean attacker from 83.6% to 2.1%, and neither the head arrangement nor the training objective moves the result.

How hard would the benchmark have to be? If the published table cannot separate the proposed model from the rule-based comparators because the benchmark is saturated, how much harder would it have to be? The generator's camouflage factor is the obvious knob, and what it does is not the intuitive thing: the factor mixes a malicious node's attack profile with an evasive reference, so a larger factor is a quieter attacker, and the effect size that matters is the one on the window summary, because a window averages the observation noise away. Moving the factor to its quietest level takes that effect size from 1.55 to 0.36, below the threshold at which an effect is conventionally called negligible, and the marginal block's held-out separability still does not fall below 93%. The margin is not only a level summary: it

also carries the recent mean, the slope, the volatility and the history channel, and a periodic attacker's intermittency is visible in all of them, so removing the level does not remove the attacker. The second knob, the benign population's own spread, which the camouflage factor does not touch at all, reaches 0.49 and 91.7%, and the two together reach 0.36 and 92.8%.

A benchmark of this family therefore does not run out of separability by being made quieter; it runs out only when the attacker's evidence is not in its own channels at all, which is the case the spatial scenario of Section 4.6 constructs and the only place in this manuscript where the method groups separate. The measurements above are a diagnosis of where the comparison can and cannot be made, not a claim that a harder benchmark would favour the proposed model.

The released material is nevertheless useful as an auditable starting point. A defensible next study should replace the synthetic allocation rules with a common policy for all detectors, measure actual runtime and messages, implement genuine multi-point observations and aggregation, and run the packet-trace study that the replication package carries, with attack parameters held out from training and with a scenario whose attacker is level-indistinguishable, so that the two blind-spot claims are tested on packet data rather than only on generated streams. The comparison this paper does not make is between two learners on the identical feature vector, which would separate the representation from the network as cleanly as the block ablation separates the blocks from one another; the replication package carries the trained comparators that such a study would need. A comparison with a learned policy, and the environment interaction and reward-defined returns such a comparison would need, would establish whether the allocation result carries over to a controller that optimises the objective rather than applying a fixed rule. Until those experiments are available, S-TIM should be described as a supervised trust-stream proof of concept, and its claims are limited to the detection and allocation results reported here.

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All authors contributed to the conceptualization, methodology, analysis, and manuscript preparation. Author roles will be specified in the camera-ready version.

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Data Availability Statement

The replication package accompanying this manuscript provides the code, configurations, trained models, evaluation outputs, statistical summaries, generated figures and baseline reimplementations needed to reproduce the trust-layer results reported here. The package also carries the packet-trace material—the single short trace, the five multi-seed traces, their per-trace metrics and regime verdicts, the converter and the simulator build note—so that the external check can be run, although the results it would give are not reported here.

Conflicts of Interest

The authors declare no conflict of interest.

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